The Goldbach–Chen Frontier, Experimentally: A Singular-Series Cancellation and the Finite-Range Half-Power Artifact

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Abstract

The binary Goldbach conjecture asks whether every even $N > 2$ is a sum of two primes; Chen's theorem replaces the second prime by a prime or semiprime. We study how the Goldbach count $R_1(N)$ (prime q) and the semiprime Chen-surplus count $R_2(N)$ respond to the Hardy–Littlewood singular series $\mathfrak{S}(N)$. Decomposing the semiprime channel $q = rs$ by its small prime factor r, each channel carries its own two-form singular series, an unconditional Euler-product identity, so the Goldbach singular series *cancels* in the ratio of Hardy–Littlewood main terms, leaving a Mertens channel sum $W_f(N) \sim \log\log N$ (hence $R_2/R_1 = W_f(N)(1+o(1))$ under the stated major-arc input). Thus $\mathfrak{S}(N)$ enters R_2 with exponent 1, exactly as in R_1, and the apparent "half power" of finite-range log–log fits is not a structural exponent but an effective exponent inferred from the regression slope $m_2 = \beta_2 - 1$, obeying $\beta_2(N) = 1 - c(N)/\langle R_2/R_1\rangle \to 1$ with deficit $\sim 1/\log\log N$. A Vaughan–Parseval minor-arc second moment (no sieve) reduces the *linear* almost-all asymptotic $R_2(N) = \mathrm{main}(N)(1+o(1))$ to the standard Hardy–Littlewood major-arc input for the channel forms $s, N - rs$; the passage to the untruncated log-regression statement $\beta_2 \to 1$ is isolated as an explicit lower-tail hypothesis. The goal is not a proof of Goldbach but an explicit cancellation identity, reproducible evidence, and an honestly bounded reduction. Additional computational reframings are summarized in an appendix and developed in the supplement.

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1 Introduction

The binary Goldbach conjecture states that every even integer $N > 2$ can be written as the sum of two primes. The conjecture remains open. However, several nearby facts are known. Helfgott proved the ternary, or weak, Goldbach conjecture: every odd integer greater than 5 is the sum of three primes [4]. Oliveira e Silva, Herzog and Pardi verified the binary Goldbach conjecture computationally up to $4 \cdot 10^{18}$ [9]. Chen's theorem proves that every sufficiently large even integer can be written as

$$N = p + q,$$

where p is prime and q is either prime or the product of at most two primes [1]; the semiprime channel and the "E_2" numbers it counts are classical sieve objects [2], and for the broader additive-bases context we refer to Nathanson [8]. Recent sieve-theoretic work also studies intermediate statements between Chen's theorem and the binary Goldbach conjecture [6].

This project focuses on the transition between the Goldbach representation

$$N = p + \text{prime}$$

and the Chen representation

$$N = p + \text{semiprime}.$$

The guiding question is not merely whether a representation exists, but how the available representations are distributed, how often the second term is genuinely prime, how often it is semiprime, and whether a representation can be selected in a smooth way as N varies.

Main idea. For each even N, define the number of representations with q prime and with q semiprime. Then study the empirical object

$$N \longmapsto \{(p, q) : p + q = N,\ p \in \mathbb{P},\ q \in \mathbb{P} \cup \mathcal{S}_2\}.$$

This is a discrete object, so it is not continuous in the ordinary analytic sense. Nevertheless, one can ask whether the induced empirical measures converge weakly, whether normalized counts have stable asymptotics, and whether a mixed-integer model can select one representation per even integer with small variation. This gives a precise mathematical language for the intuition that the representation may "jump" between prime and semiprime behavior.

Summary of findings. Sections 7–7.7 report the results of implementing this program; all numbers are reproducible from the accompanying code. The main outcomes are:

1. **The singular series of R_2, and the finite-range half-power (§7.2).** Counts are computed for all $N \le 2 \cdot 10^6$ by FFT and locally in windows up to 10^9 by a block-convolution sieve. Both main terms carry $\mathfrak{S}(N)$ with exponent 1: the per-channel singular series factorizes (Lemma 1), so $\mathfrak{S}(N)$ *cancels* in the main-term ratio, leaving $R_2/R_1 = W_f(N)(1 + o(1))$ with the Mertens channel sum $W_f(N) \sim \log\log N$. The number $\beta_2 \approx \frac{1}{2}$ that a finite-range log–log fit reports is therefore not a structural exponent but a finite-range effective exponent inferred from the regression slope $m_2 = \beta_2 - 1$; a derivation and the data agree on $\beta_2(N) = 1 - c/\langle R_2/R_1\rangle_N$ with $\langle R_2/R_1\rangle \sim \log\log N$, so $\beta_2 \to 1$ with an $\mathcal{O}(1/\log\log N)$ deficit (validation $(1 - \beta_2)\langle R_2/R_1\rangle \approx 1.09$ to 1%). The deficit is multi-channel: deleting the heaviest channel ($3 \mid N$) removes only $\sim 10\%$ of it (§7.2), refuting a single-atom reading. The channel factorization itself is an elementary singular-series computation; what we record is its consequence—the cancellation of $\mathfrak{S}(N)$ in the main-term ratio—which the existence-oriented sieve literature has had no reason to isolate (Remark 3).
2. **Layers and sign reversal (§7.3).** For $\Omega(q) = k$, the effective exponent β_k falls through $1, \frac{1}{2}, \ldots$ and becomes negative for $k \ge 3$: $\mathfrak{S}(N)$ enhances low-complexity and suppresses high-complexity channels.
3. **Diagnostics, continuity, optimization (§7.4–7.7).** Factor geometry favours unbalanced semiprimes; the q/N measures stabilize weakly to a near-uniform law; minimum-variation selection is a Pareto trade-off between geometric and type smoothness; the Chen-rescue threshold under a balance constraint is unimodal; and the residual balance exponent $a_{\mathrm{Res}}(N)$ correlates positively with $\mathfrak{S}(N)$.
4. **An almost-all asymptotic, honestly bounded (Appendix A).** The cancellation identity (Lemma 1) is unconditional. A circle-method second-moment estimate—Vaughan's classical *prime* minor-arc bound plus the trivial L^2 size of the semiprime sum, no sieve—is unconditional and reduces the *linear* asymptotic $R_2(N) = \mathrm{main}(N)(1 + o(1))$ (all but $o(X)$ even N) to the standard Hardy–Littlewood major-arc evaluation for the channel forms $s, N - rs$; we present it within that stated major-arc reduction rather than as a self-contained

theorem. Upgrading to the log-regression statement $\beta_2(N) \to 1$ needs one further input (an explicit lower-tail hypothesis on R_2/R_1, beyond L^2); we state that gap rather than claim past it.

5. **Many mirrors, one fact (§8.1–8.2; Appendix B).** The frontier can be recast as a thermodynamic ground state, an adversarial network, a liquidity market, a control problem, an empirical circle method, an information channel, and a discrete surface. We are explicit that this is *one* finding—$\mathfrak{S}(N)$ sets the local densities—observed many times, not many findings; most reframings are deterministic functions of the same $\mathfrak{S}/R_2$ data and are collected in Appendix B (with the honest negatives: an empty additive-complexity topology and a Chen substitute with no resilience premium). Two earn a place in the body by yielding a *new computation*: the empirical circle method (§8.1), whose comet power spectrum is an empirical shadow of the major arcs, and the carry geometry (§8.2), which surfaces a digit/Kummer signature of primality.

Status of the results: proven, conditional, conjectural. Because the central object is an empirical exponent that is easy to mistake for a structural one, we state once, explicitly, what is proved unconditionally, what is conditional and on which input, and what is left as a precise conjecture (Table 1); the body labels each result accordingly (**Theorem/Proposition** for proved statements, **Numerical Observation** for empirical ones, **Hypothesis/Conjecture** for what we do not prove).

Statement	Status
Channel singular-series factorization $\mathfrak{S}_2(r,N) = 2C_2\mathfrak{S}(N)\frac{r-1}{r-2}$ (Lemma 1)	*Proved*, unconditional, elementary Euler-product computation
Main-term cancellation $R_{2,\mathrm{main}}/R_{1,\mathrm{main}} = W_f(N)$	*Conditional* on the stated Hardy–Littlewood / major-arc main-term input
Finite-range effective exponent $\beta_2 \approx \frac{1}{2}$ and its drift; fit (22)	*Numerical observation* (not extrapolated to a limit)
Functional form $1-\beta_2 \asymp 1/\log\log N$ (Theorem 1)	*Conditional* (quantitative Hardy–Littlewood, Assumption 1); the order is robust, the limiting constant open
Non-circular validation $(1-\beta_2)\langle R_2/R_1\rangle \approx \mathrm{const}$ (Obs. 1)	*Numerical observation* (control)
Linear almost-all $R_2 = \mathrm{main}(1+o(1))$ (Theorem 3)	*Conditional* bounded reduction to a standard major-arc input; a standard circle-method consequence (Remark 8)
$\beta_2 \to 1$ for almost all N (Corollary 1)	*Conditional* on the lower-tail hypothesis
Lower-tail hypothesis	*Conjecture*, stated precisely (§A.6)
Diagnostics of §7.4–7.8 (balance, continuity, valleys, transport, dynamics)	*Numerical observations*
Model statements in Appendix A (blocked channel, weak uniformity, least-unblocked prime, carry excess)	*Proved in the stated model*

Table 1: What is proved, what is conditional, and what is conjectured. The half-power $\beta_2 \approx \frac{1}{2}$ is a numerical observation; the structural exponent is 1 (Lemma 1).

2 Basic notation

Let $\mathbb{P}$ denote the set of primes. For an integer $n \geq 2$, let $\Omega(n)$ be the number of prime factors of n counted with multiplicity. Thus $\Omega(12) = 3$ because $12 = 2^2 \cdot 3$.

Definition 1 (Semiprimes and Chen numbers)**.** *Define*

$$\mathcal{S}_2 := \{n \geq 2 : \Omega(n) = 2\}$$

to be the set of semiprimes, allowing squares of primes. Define also

$$\mathcal{P}_2 := \{n \geq 2 : \Omega(n) \leq 2\} = \mathbb{P} \cup \mathcal{S}_2.$$

Thus $\mathcal{P}_2$ is the set of integers with at most two prime factors counted with multiplicity.

For an even integer N, define the prime–prime representation count

$$R_1(N) := \sum_{p<N} \mathbf{1}_{\mathbb{P}}(p)\mathbf{1}_{\mathbb{P}}(N-p). \tag{1}$$

This counts ordered representations by choosing the first summand p. If unordered counts are desired, one may restrict to $p \leq N/2$.

Define the prime–semiprime representation count

$$R_2(N) := \sum_{p<N} \mathbf{1}_{\mathbb{P}}(p)\mathbf{1}_{\mathcal{S}_2}(N-p). \tag{2}$$

Finally, define the Chen count

$$R_{\leq 2}(N) := R_1(N) + R_2(N) = \sum_{p<N} \mathbf{1}_{\mathbb{P}}(p)\mathbf{1}_{\mathcal{P}_2}(N-p). \tag{3}$$

Remark 1. *The binary Goldbach conjecture is equivalent to $R_1(N) > 0$ for every even $N > 2$. Chen's theorem implies $R_{\leq 2}(N) > 0$ for all sufficiently large even N.*

3 The Goldbach–Chen frontier

The central empirical object is the decomposition of Chen representations into genuinely Goldbach and genuinely semiprime representations:

$$R_{\leq 2}(N) = R_1(N) + R_2(N).$$

This suggests several diagnostics.

3.1 Chen surplus and Goldbach fragility

Definition 2 (Chen surplus)**.** *For even N, define the Chen surplus ratio*

$$C(N) := \frac{R_2(N)}{R_1(N)+1}. \tag{4}$$

Large values of $C(N)$ identify integers for which semiprime representations are much more abundant than prime–prime representations.

Definition 3 (Goldbach fragility)**.** *For a threshold $k \in \mathbb{N}$, define the set of k-fragile even integers up to X by*

$$\mathcal{F}_k(X) := \{N \leq X : N \equiv 0 \pmod 2,\ R_1(N) \leq k,\ R_2(N) > 0\}. \tag{5}$$

These are integers for which Goldbach representations exist only sparsely, while Chen-type representations remain available.

Research question. How large is $\mathcal{F}_k(X)$? Are fragile integers concentrated in specific congruence classes? Does fragility correlate with the singular series correction in the Hardy–Littlewood heuristic?

3.2 Prime versus semiprime share

Define

$$\theta(N) := \frac{R_1(N)}{R_1(N) + R_2(N)}, \tag{6}$$

with the convention $\theta(N) = 0$ if the denominator is zero.

Here $\theta(N)$ is the empirical share of Chen representations in which q is prime rather than semiprime. A small value of $\theta(N)$ means that most Chen representations use a genuinely semiprime second summand.

A naive density heuristic suggests

$$R_1(N) \asymp \frac{N}{\log^2 N}, \qquad R_2(N) \asymp \frac{N \log\log N}{\log^2 N},$$

because primes have local density roughly $1/\log N$, while semiprimes have density roughly $\log\log N/\log N$. Therefore, at a first heuristic level, semiprime representations should outnumber prime–prime representations by a factor of order $\log\log N$. This does not prove Goldbach or Chen, but it motivates the following empirical conjecture.

Conjecture 1 (Average semiprime dominance)**.** *There exists a slowly varying normalization $L(X) \sim \log\log X$ such that*

$$\frac{1}{X/2} \sum_{\substack{N \le X \\ N \equiv 0\,(2)}} \frac{R_2(N)}{R_1(N) + 1} \asymp L(X).$$

Equivalently, prime–semiprime representations should become increasingly abundant relative to prime–prime representations on average.

4 Distribution of the second summand q

For each even N, define the representation sets

$$\mathcal{R}_1(N) := \{(p, q) : p + q = N,\ p \in \mathbb{P},\ q \in \mathbb{P}\}, \tag{7}$$

$$\mathcal{R}_2(N) := \{(p, q) : p + q = N,\ p \in \mathbb{P},\ q \in \mathcal{S}_2\}, \tag{8}$$

$$\mathcal{R}_{\le 2}(N) := \mathcal{R}_1(N) \cup \mathcal{R}_2(N). \tag{9}$$

The object $q = N - p$ can be studied in several ways.

4.1 Location of q inside $[0, N]$

Define empirical measures on $[0, 1]$:

$$\mu_N^{(1)} := \frac{1}{R_1(N)} \sum_{(p,q) \in \mathcal{R}_1(N)} \delta_{q/N}, \tag{10}$$

$$\mu_N^{(2)} := \frac{1}{R_2(N)} \sum_{(p,q) \in \mathcal{R}_2(N)} \delta_{q/N}, \tag{11}$$

$$\mu_N^{(\le 2)} := \frac{1}{R_{\le 2}(N)} \sum_{(p,q) \in \mathcal{R}_{\le 2}(N)} \delta_{q/N}, \tag{12}$$

whenever the denominators are nonzero.

Question 1 (Weak continuity). *Do the measures $\mu_N^{(1)}$, $\mu_N^{(2)}$, or $\mu_N^{(\le 2)}$ converge weakly along typical even N? If so, is the limit close to the uniform distribution on $(0,1)$ after removing small endpoint effects?*

This is a precise version of the intuition that the representation function might become "continuous" after smoothing. The pointwise object is discontinuous, because primality is discrete; but the normalized cloud of representations may stabilize in distribution.

4.2 Type-labeled empirical measures

Let the type of q be

$$\tau(q) := \begin{cases} 1, & q \in \mathbb{P}, \\ 2, & q \in \mathcal{S}_2. \end{cases}$$

Define a type-labeled empirical measure on $[0,1] \times \{1,2\}$ by

$$\nu_N := \frac{1}{R_{\le 2}(N)} \sum_{(p,q)\in\mathcal{R}_{\le 2}(N)} \delta_{(q/N,\tau(q))}. \tag{13}$$

Conjecture 2 (Weak type stabilization). *For typical large even N, the measures ν_N are close to a product-like law*

$$\nu_N \approx f_N(x,t)\, dx\, d\tau,$$

where the spatial component in $x = q/N$ varies slowly, while the type marginal increasingly favors $t = 2$ because semiprimes have larger density than primes.

This is not ordinary continuity. It is closer to convergence of empirical distributions, or continuity after smoothing over windows $N \in [X, X+H]$.

4.3 Factor geometry of semiprime q

If $q \in \mathcal{S}_2$, write

$$q = ab, \qquad a, b \in \mathbb{P}, \qquad a \le b.$$

Then define the balance statistic

$$B(q) := \frac{\log a}{\log q} \in \left(0, \frac{1}{2}\right]. \tag{14}$$

A value near $1/2$ means that q is a balanced semiprime; a value near 0 means that q has a very small prime factor.

Question 2 (Semiprime factor geometry). *Among representations $N = p + q$ with $q \in \mathcal{S}_2$, how is $B(q)$ distributed? Are Chen representations dominated by unbalanced semiprimes with a small factor, or by balanced semiprimes?*

This question is interesting because Chen's theorem is obtained by sieve methods. Empirically, one can ask whether the extra freedom in q comes mostly from allowing small prime divisors, or whether semiprime representations are broadly distributed across factor shapes.

5 Heuristic asymptotics (classical recovery)

The asymptotics in this section are classical; we recall them only to fix normalisation and as a control for the computations of §7, not as new derivations. The Hardy–Littlewood circle-method heuristic for Goldbach [3] predicts a main term of the form

$$R_1(N) \approx 2C_2\, \mathfrak{S}(N) \frac{N}{\log^2 N}, \tag{15}$$

where C_2 is the twin-prime constant and $\mathfrak{S}(N) = \prod_{p|N,\, p>2} \frac{p-1}{p-2}$ is the singular-series correction depending on the odd prime divisors of N (the factor $2C_2$ already absorbs the $p = 2$ local density).

For semiprime q, Landau's density of numbers with two prime factors [5], $\#\{n \le x : \Omega(n) = 2\} \sim x \log\log x / \log x$, gives the rough estimate

$$\#\{n \le x : \Omega(n) = 2\} \sim \frac{x \log\log x}{\log x}. \tag{16}$$

Therefore, a first-order independence heuristic gives

$$R_2(N) \approx K_2(N) \frac{N \log\log N}{\log^2 N}, \tag{17}$$

where $K_2(N)$ is a correction factor capturing congruence constraints and local arithmetic structure.

Normalization. The empirical project should therefore compare

$$A_1(N) := \frac{R_1(N)}{N/\log^2 N}, \tag{18}$$

$$A_2(N) := \frac{R_2(N)}{N \log\log N/\log^2 N}. \tag{19}$$

If the heuristic is meaningful, $A_1(N)$ and $A_2(N)$ should be much more stable than the raw counts.

6 Selection formulations

Scope note. This section records the optimization *language* used by two diagnostics measured in §7.5 (a minimum-variation selection and a Chen-rescue threshold). It is deliberately short: the feasibility version is trivial and no MIP solver will prove Goldbach (Remark 2). The infinite-dimensional and column-generation readings are recorded in the supplement; a reader after the empirical results may skip to §7.

Fix an even $N \le X$ and let $\mathcal{A}_N^{(1)} = \{p \in \mathbb{P} : N - p \in \mathbb{P}\}$ and $\mathcal{A}_N^{(2)} = \{p \in \mathbb{P} : N - p \in \mathcal{S}_2\}$. Binary variables $z_{N,p,t}$ select one representation $N = p + q$ of type t (prime $t = 1$, semiprime $t = 2$); the cover constraint $\sum_{p \in \mathcal{A}_N^{(1)}} z_{N,p,1} + \sum_{p \in \mathcal{A}_N^{(2)}} z_{N,p,2} \ge 1$ for every even N in a window $\mathcal{E}_X$ is Chen feasibility, and dropping the type-2 terms is Goldbach feasibility. With the sets $\mathcal{A}_N^{(t)}$ precomputed by a sieve this is trivial as a feasibility problem; it becomes informative only under added structure.

Remark 2 (Why this is not a proof strategy)**.** *These mixed-integer programs do not make Goldbach easier by themselves. They translate the conjecture into a covering problem over a hypergraph whose vertices are even integers and whose hyperedges are prime-pair certificates; the value is conceptual, letting one import selection, smoothness, total variation, and column generation into the empirical study of Goldbach-type representations (developed in the supplement).*

Minimum-variation selection. Selecting one representation per even N in a window, with type $u_N = \sum_{p \in \mathcal{A}_N^{(2)}} z_{N,p,2} \in \{0, 1\}$ and normalized location $s_N = q_N/N = \sum_{p,t} \frac{N-p}{N} z_{N,p,t}$ (linear in the selection variables), the minimum-variation MIP minimizes $\alpha \sum_N v_N + \beta \sum_N r_N + \gamma \sum_N |s_N - \frac{1}{2}|$ subject to one selection per N and $v_N \ge |u_N - u_{N-2}|$, $r_N \ge |s_N - s_{N-2}|$. The term $\sum v_N$ measures type oscillation, $\sum r_N$ geometric oscillation, and $|s_N - \frac{1}{2}|$ rewards balance. The guiding question is whether the minimum type-switch rate $|\mathcal{E}_X|^{-1} \min \sum_N v_N$ decreases or stays bounded away from 0 as $X \to \infty$; §7.5 reports the answer (a smooth-location branch that nonetheless keeps switching type).

Chen-rescue threshold. Restricting representations to a balanced band $p \in [N/2 - k,\, N/2]$ and counting how many N *require* a semiprime (no prime pair in the band) defines the rescue fraction, whose unimodal profile in k is reported in §7.5; if the minimum number of semiprime selections is zero over a finite range, Goldbach holds there.

7 Empirical results and the singular-series regularity

This section reports the outcome of implementing the program above. All numbers are reproducible from the accompanying code (see the Data and code availability statement; the drivers `run_counts.py`, `run_scaling.py`, `run_betalaw.py`, `run_estimator_audit.py`); unless stated otherwise they use the range $X = 2 \cdot 10^6$.

7.1 Computational setup and validation

The counts R_1 and R_2 are obtained for *all* $N \le X$ at once as additive convolutions,

$$R_1 = \mathbf{1}_{\mathbb{P}} * \mathbf{1}_{\mathbb{P}}, \qquad R_2 = \mathbf{1}_{\mathbb{P}} * \mathbf{1}_{\mathcal{S}_2},$$

evaluated by FFT in $O(X \log X)$. For $X = 2 \cdot 10^6$ this takes about 15 seconds including sieves, diagnostics and figures. The FFT counts agree exactly with a brute-force enumeration on random samples (the rounding error of the transform is far below 1/2), and Goldbach's condition $R_1(N) > 0$ is verified throughout the range.

7.2 The semiprime channel and the finite-range half-power

The Hardy–Littlewood prediction (15) attaches to R_1 the full singular series $\mathfrak{S}(N) = \prod_{p|N,\, p>2}(p-1)/(p-2)$. Regressing $\log(R_1/(N/\log^2 N))$ against $\log \mathfrak{S}(N)$ recovers, empirically, a slope 1.000, and dividing by $\mathfrak{S}(N)$ collapses the coefficient of variation of the normalized count from 0.39 to 0.015 (a 25× reduction); this is a control that reproduces (15) and the band structure of the Goldbach comet.

The channel observation concerns R_2. We measure how strongly the relative semiprime share tracks $\mathfrak{S}(N)$ through the regression of $\log(R_2/R_1)$ on $\log \mathfrak{S}(N)$, which uses R_1 itself as a meter for $\mathfrak{S}(N)$ and is therefore independent of any chosen normalization.

Regression notation. To avoid confusing the raw regression slope with the effective singular exponent, we write

$$m_2(X) := \mathrm{slope}_{5000 \le N \le X} \left(\log \frac{R_2(N)}{R_1(N)} \ \text{ on } \ \log \mathfrak{S}(N) \right), \qquad \beta_2(X) := 1 + m_2(X).$$

Thus $\beta_2 = 1$ means that R_2 carries the same singular-series power as R_1, while $m_2 = 0$ means that the ratio R_2/R_1 is uncorrelated with $\mathfrak{S}(N)$. We call β_2 the *effective singular exponent* of R_2: it is an effective exponent inferred from a regression slope, and the raw slope is $m_2 = \beta_2 - 1$. As Lemma 1 shows below, $\mathfrak{S}(N)$ enters R_2 with exponent 1 (exactly as in R_1) and *cancels* in the ratio R_2/R_1, so R_2/R_1 is not a power of $\mathfrak{S}$ and the value $\beta_2 \approx \frac{1}{2}$ reported in finite ranges is a finite-size correction, not a structural exponent (developed after Table 2). All tables below report β_2, not the raw slope m_2, using one estimator throughout: ordinary least squares, cumulative over all even $5000 \le N \le X$. The reported OLS standard errors are descriptive only, since consecutive even integers are arithmetically dependent; a block-bootstrap robustness check is left as a diagnostic in the replication code, and the present tables report descriptive OLS uncertainty only. The appendix (§A.6) instead uses a *local* window centred at X, which runs ≈ 0.02–0.03 higher at the same X—the distinction is between an average over $[5000, X]$ and a measurement at the leading edge.

X	β_2	R^2_{OLS} of the regression
$5 \cdot 10^5$	0.452	0.91
$1 \cdot 10^6$	0.475	0.91
$2 \cdot 10^6$	0.497	0.92
$4 \cdot 10^6$	0.513	0.93
$8 \cdot 10^6$	0.529	0.93

Table 2: Effective singular exponent $\beta_2 = 1 + m_2$ of R_2, where m_2 is the cumulative OLS slope of $\log(R_2/R_1)$ on $\log \mathfrak{S}$ over $[5000, X]$; the table reports β_2, not the raw slope $m_2 = \beta_2 - 1$. The value sits near $\frac{1}{2}$ in this range and *rises* with X—already a warning that it is not a fixed exponent (contrast $\beta_1 = 1.000$, which is genuinely scale-invariant). The singular series explains about 92% of the variance of $\log(R_2/R_1)$. OLS standard errors are descriptive only (nearby N are arithmetically dependent).

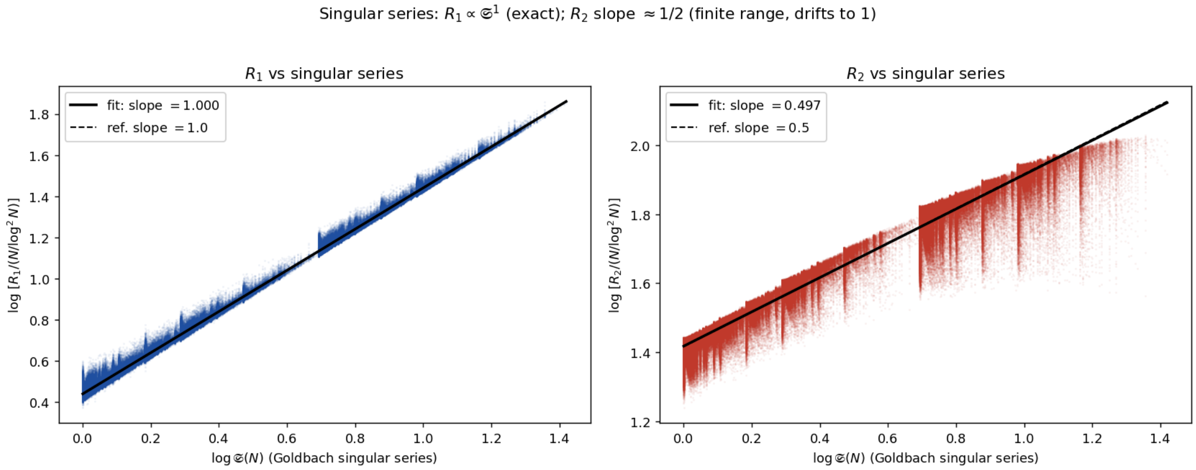

Figure 1: Left: $\log[R_1/(N/\log^2 N)]$ against $\log \mathfrak{S}(N)$ has slope 1.000 (Hardy–Littlewood). Right: the same regression for R_2 gives effective exponent $\beta_2 = 0.497$ at $X = 2 \cdot 10^6$, near the reference line $\beta = 1/2$—but this is the *finite-range* effective exponent of a non-power-law ratio (it drifts toward 1; Fig. 2), not a fixed power. The vertical striations are the discrete values of $\mathfrak{S}(N)$.

Over this range, the best finite-range log–log normalization is consistent with the effective fit (near $X \sim 10^6$, the effective exponent $\frac{1}{2}$ drifting to 1; §7.2 below)

$$R_2(N) \;\propto\; \mathfrak{S}(N)^{1/2}\,\frac{N}{\log^2 N}\,W_0(N), \qquad W_0(N) = \sum_{\substack{3 \le r \le \sqrt{N} \\ r \text{ prime}}} \frac{r-1}{r-2}\,\frac{1}{r}\,\frac{\log N}{\log(N/r)}, \tag{20}$$

with $W_0(N) \sim \log\log N$ by Mertens' theorem [12]; here W_0 is the *unblocked* Mertens channel sum, ranging over all small primes r. The theoretically relevant object is instead the *blocked* channel sum W_f, which omits the channels $r \mid N$ (where $p = N - rs \equiv 0 \bmod r$ forces $p = r$),

$$W_f(N) = \sum_{\substack{3 \le r \le \sqrt{N} \\ r \in \mathbb{P},\ r \nmid N}} \frac{r-1}{r-2}\,\frac{1}{r}\,\frac{\log N}{\log(N/r)}, \tag{21}$$

the same sum W_f that governs the appendix derivation, where it is the channel-cancellation factor $R_2/R_1 = W_f(N)(1 + o(1))$ (Theorem 1, conditional). A notational caveat: in (20), (21) and Table 3 the channel weight is the *elementary* back-of-envelope form $w_r \approx \frac{1}{r}\frac{\log N}{\log(N/r)}$ displayed above, whereas the appendix and Theorem 3 use the *exact* archimedean weight

$w_r = I_r(N)/I_1(N)$ that it approximates; the elementary form overstates I_r/I_1 by a constant factor for small r (quantified numerically in §A.12), which shifts the constants but not the qualitative comparison. We use the elementary weight in Table 3 for comparability with the finite-range fit (20); recomputing the four dispersions with the exact weight I_r/I_1 on a subsample changes the constants only slightly (0.184/0.167/0.019/0.0135 → 0.184/0.167/0.019/0.016) and leaves the ordering—and the blocked full-inheritance normalization as the tightest—unchanged. Table 3 compares the dispersion of R_2 under four normalizations.

normalization of R_2	CV ($X = 2 \cdot 10^6$)
$N \log\log N/\log^2 N$ (heuristic of §5)	0.184
$2C_2\,\mathfrak{S}(N)\,N\,W_0/\log^2 N$ (full $\mathfrak{S}$, *unblocked* W_0)	0.167
$2C_2\,\mathfrak{S}(N)^{1/2}\,N\,W_0/\log^2 N$ (half power, W_0)	0.019
$2C_2\,\mathfrak{S}(N)\,N\,W_f/\log^2 N$ (**full inheritance, blocked channels**)	**0.0135**

Table 3: Dispersion of R_2 under four normalizations. The finite-range half-power normalization is descriptive rather than structural: it fits tightly (CV 0.019) only because it pairs the unblocked Mertens sum W_0 with the *wrong* power $\mathfrak{S}^{1/2}$. The theoretically relevant full-inheritance normalization is the *blocked*-channel expression $2C_2\mathfrak{S}(N)W_f(N)N/\log^2 N$, which carries the *full* singular series ($\beta = 1$) and fits *tighter still* (CV 0.0135); forcing the full series with the *unblocked* W_0 instead fits worst (0.167). The apparent 1/2 power thus summarizes, over a finite range, the covariance between the blocked channels $r \mid N$ and $\mathfrak{S}(N)$; it is not a limiting singular-series exponent.

Interpretation: a quantitative Chen rescue effect. Because the finite-range effective exponent satisfies $\beta_2 < 1$, over this range the relative semiprime share falls with the singular series, $R_2/R_1 \propto \mathfrak{S}(N)^{-(1-\beta_2)}$ (locally $\approx \mathfrak{S}(N)^{-1/2}$): where Goldbach is locally rich (large $\mathfrak{S}$), the prime channel dominates, whereas where Goldbach is *fragile* (small $\mathfrak{S}$) the semiprime channel is proportionally most abundant. This makes the Chen-rescue intuition—the semiprime channel matters most exactly where primes are scarce—quantitative as an *effective finite-window scaling*, with the deficit $1-\beta_2 \asymp 1/\log\log N$ vanishing asymptotically (so β_2 is an effective exponent, not a fixed power; §7.2 opening). Heuristically the effective exponent sits below 1 because writing $q = rs$ couples the divisors of N to the factor r: when a small prime $\ell \mid N$, the channel $r = \ell$ is *blocked* (then $p = N - \ell s \equiv 0 \bmod \ell$ forces $p = \ell$), removing its term from the sum over channels precisely when $\mathfrak{S}(N)$ is large (Lemma 1). This explains the direction $\beta_2 < 1$; the precise value is not 1/2, as the scaling analysis below shows.

Scaling to 10^9: the effective exponent drifts up like $\log\log X$. Measuring β_2 *locally* in a window of $2 \cdot 10^6$ even integers centred at X (computed by a block-convolution sieve, so that X can reach 10^9 while only the window is held in memory), the effective exponent does *not* settle at 1/2: it rises monotonically (Fig. 2). Eight windows from $6 \cdot 10^6$ to 10^9 are fit to within 0.0013 by

$$\beta_2(X) \;\approx\; -0.094(7) + 0.234(3)\,\log\log X \qquad (R^2_{\mathrm{OLS}} = 0.9987,\ 8\text{ windows}), \tag{22}$$

the parenthesised figures being the last-digit standard errors of the two-parameter fit. Thus the half-power $\beta_2 \approx \frac{1}{2}$ is only the *local* value near $X \sim 10^6$, and the effective exponent is not a constant. The fit reaches $\beta_2 \approx 0.61$ at 10^9 (measured). We deliberately *do not* read a numerical limit off (22): extrapolating a $\log\log X$ trend fitted over three decades to, say, $X \sim 10^{46}$ (where it would touch $\beta_2 = 1$) is well outside the data and is not evidence—the robust content is the *functional form* $1 - \beta_2 \asymp 1/\log\log X$ and its derivation below, not a forecast of where the effective exponent crosses a particular value. The regression quality stays high throughout ($R^2_{\mathrm{OLS}} \approx 0.997$, CV of the half-power normalization $\lesssim 0.04$).

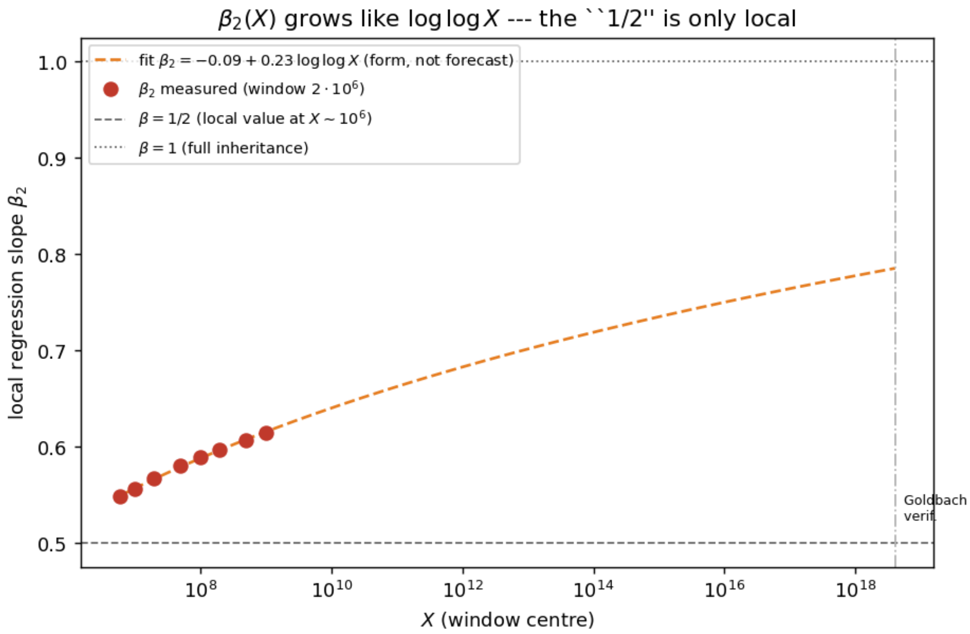

Figure 2: Local effective singular exponent $\beta_2(X) = 1 + m_2(X)$ for windows up to 10^9, with the $\log\log X$ fit (22) (the dashed extrapolation beyond 10^9 is shown only to display the functional form, not as a prediction). The effective exponent drifts steadily above 1/2 without sign of leveling off, consistent with $\beta_2 \to 1$.

A derivation: $\beta_2 \to 1$ with an $1/\log\log N$ deficit. The drift is not arbitrary. Writing $q = N - p = rs$ and summing over the small factor r, each channel is a Goldbach count for the linear form $(s, N - rs)$, whose two-form singular series factorizes as

$$\mathfrak{S}_2(r, N) = \begin{cases} 2C_2\, \mathfrak{S}(N)\, \dfrac{r-1}{r-2}, & r \nmid N, \\ 0, & r \mid N \end{cases} \tag{23}$$

(Lemma 1; the $r \mid N$ channel is identically blocked). The Hardy–Littlewood main terms are $R_{1,\text{main}} = 2C_2\mathfrak{S}(N)I_1$ and $R_{2,\text{main}} = 2C_2\mathfrak{S}(N)\sum_{r\nmid N} \frac{r-1}{r-2} I_r$, so the singular series *cancels exactly* between them,

$$\frac{R_{2,\text{main}}(N)}{R_{1,\text{main}}(N)} = W_f(N) := \sum_{\substack{3 \le r \le \sqrt{N} \\ r \nmid N}} \frac{r-1}{r-2}\, w_r(N), \qquad w_r = \frac{1}{r}\frac{\log N}{\log(N/r)}, \tag{24}$$

and hence $R_2(N)/R_1(N) = W_f(N)(1 + o(1))$ for the actual counts under the per-channel asymptotic. Hence the raw regression slope is $m_2 = \beta_2 - 1 = \operatorname{Cov}(\log W_f, \log \mathfrak{S}) / \operatorname{Var}(\log \mathfrak{S})$, and the only coupling to $\mathfrak{S}(N)$ comes from the blocked channels $r \mid N$, each *removed* from W_f. Modelling $\mathbf{1}[\ell \mid N]$ as independent with probability $1/\ell$, $\operatorname{Var}(\log \mathfrak{S}) =: B$ is a convergent constant, while the removed terms contribute $\operatorname{Cov}(\log W_f, \log \mathfrak{S}) = -A/W_{f,0}$ with $A > 0$ constant and $W_{f,0} \approx \langle R_2/R_1 \rangle \sim \log\log N$ the baseline sum (Theorem 1). Therefore

$$\beta_2(N) = 1 - \frac{c(N)}{\langle R_2/R_1 \rangle_N} + o\!\left(\frac{1}{\log\log N}\right), \qquad c(N) = \tfrac{A}{B} = O(1)\ (\approx 1.1 \text{ in range; Fig. 15}), \tag{25}$$

so $\beta_2 \to 1$: R_2 *asymptotically inherits the full Goldbach singular series*, the approach being governed by $1/\log\log N$. This explains why finite ranges show $\beta_2 \approx \frac{1}{2}$ and a slow apparent drift.

Numerical Observation 1 (Non-circular validation)**.** *The law* (25) *predicts that* $(1-\beta_2)\langle R_2/R_1\rangle$ *is constant. Measured over windows from* $6\cdot10^6$ *to* $2\cdot10^8$ *it equals* $1.11, 1.11, 1.10, 1.09, 1.09, 1.08$*, a coefficient of variation of* 1% *(versus* 1.7% *for* $(1-\beta_2)\log\log N$*): the correct "clock" is* $\langle R_2/R_1\rangle$*, exactly as* (25) *requires (Fig. 3). This is non-circular because* β_2 *(a covariance/variance slope) and* $\langle R_2/R_1\rangle$ *(a mean) are* a priori *independent statistics.*

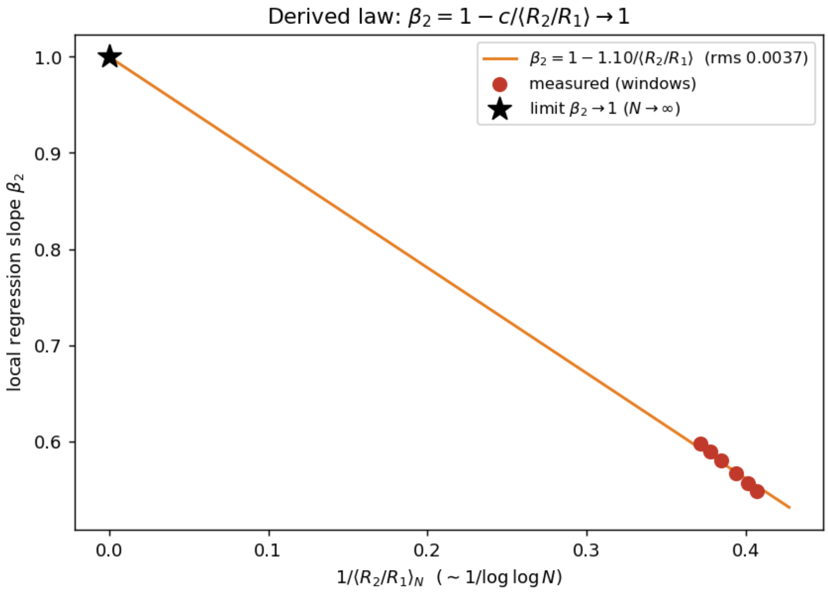

Figure 3: The derived law: β_2 against $1/\langle R_2/R_1\rangle \sim 1/\log\log N$ is a straight line through the limit point $(0,1)$, i.e. $\beta_2 \to 1$.

Not a single-atom effect. A natural worry is that the whole deficit $1-\beta_2$ is the arithmetic of one prime: $3 \mid N$ simultaneously doubles $\mathfrak{S}(N)$ (the factor $\frac{3-1}{3-2} = 2$) and deletes the heaviest semiprime channel $r = 3$, so one might suspect $\beta_2 \approx \frac{1}{2}$ is just "what divisibility by 3 does" near $X \sim 10^6$. The data refute this. Recomputing β_2 on the subpopulation $3 \nmid N$ (the $r = 3$ atom removed) gives, at $X = 2\cdot10^6$, $\beta_2 = 0.547 \pm 0.0004$ against 0.495 on all N: the deficit $1-\beta_2$ falls by only about 10% (`run_estimator_audit.py`). The mechanism is real—conditional means confirm $\langle\log\mathfrak{S} \mid 3 \mid N\rangle - \langle\log\mathfrak{S} \mid 3 \nmid N\rangle = 0.69 \approx \log 2$ and a depressed R_2/R_1 on $3 \mid N$—but it accounts for a tenth of the deficit, not the bulk. The negativity of $\beta_2 - 1$ is genuinely a *sum over channels* $r = 3, 5, 7, \ldots$, each blocked on the sparse set $r \mid N$, exactly as $A(X) = \sum_r(\cdots)$ in (25); it is not the signature of a single residue class.

7.3 The singular exponent across arithmetic-complexity layers

The decomposition R_1, R_2 extends to layers $R_k(N) = \#\{p < N : \Omega(N-p) = k\}$, counting representations whose second summand has exactly k prime factors (Goldbach is $k = 1$, Chen's semiprime channel is $k = 2$). Measuring the singular exponent β_k of each layer – the slope of $\log(R_k/R_1)$ against $\log\mathfrak{S}(N)$, plus one – reveals a clean pattern (Table 4, Fig. 4):

Sign reversal of the singular response. The Goldbach singular series *enhances* the low-complexity channels ($\beta_1 = 1$, $\beta_2 \approx \frac{1}{2} > 0$) but *suppresses* the high-complexity ones ($\beta_k < 0$ for $k \geq 3$): where $\mathfrak{S}(N)$ is large, representations with a many-factored q become rarer. The mechanism is the same divisor effect seen for R_2, now run in reverse. When $\mathfrak{S}(N)$ is large, N is divisible by many small primes; the admissible p avoid those residues, which forces $q = N - p$ to

k	1	2	3	4	5	6
share $\rho(k)$	15.5%	33.5%	28.7%	14.6%	5.5%	1.8%
β_k	1.00	0.50	-0.30	-1.38	-2.73	-4.04

Table 4: Layer profile and singular exponent at $X = 2 \cdot 10^6$ (each regression has $R^2_{\mathrm{OLS}} \approx 0.92$). The exponent decreases monotonically and crosses zero between $k = 2$ and $k = 3$; it is *not* $1/k$.

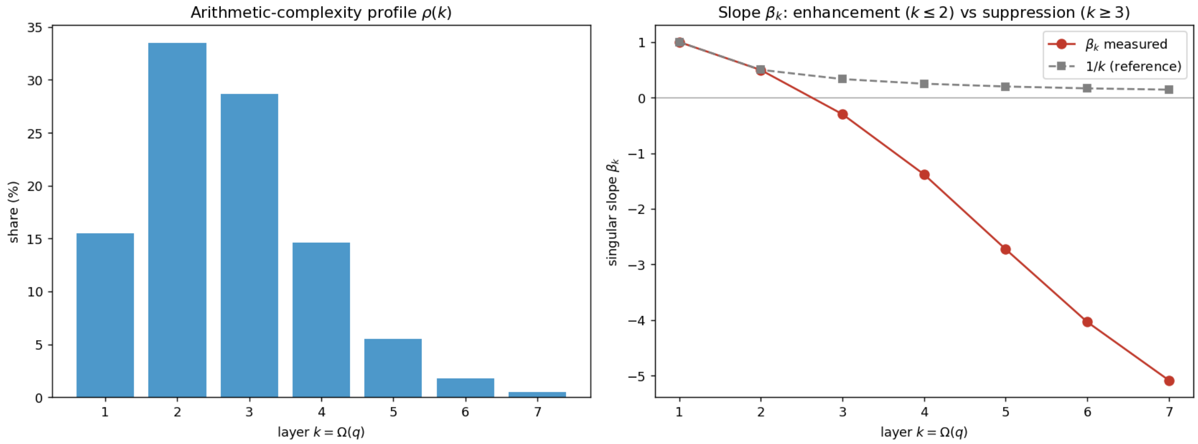

Figure 4: Left: the arithmetic-complexity profile $\rho(k)$ peaks at $k = 2$. Right: the singular exponent β_k decreases through $1, \frac{1}{2}, \ldots$ and becomes negative for $k \geq 3$, diverging from the naive $1/k$ reference.

also avoid small prime factors, so q tends to have *fewer* prime factors. Thus a large singular series biases q toward low arithmetic complexity. All β_k drift upward with X at the same slow rate as β_2 (e.g. β_3 moves from -0.30 at $2 \cdot 10^6$ to -0.20 at $8 \cdot 10^6$), so the crossing point of $\beta_k = 0$ moves slowly to the right. We also find that the strongest "rescue" channel, $q = \text{composite} \times \text{composite}$ (both factors composite), is available for *every* even N in the range with on average $\sim 1.8 \cdot 10^4$ representations: the additive hierarchy is extremely redundant below the prime layer. The whole sign reversal is the coefficient-level shadow of a single generating-function response in the blocked-channel model, made precise in Appendix A.8.

7.4 Diagnostics

The prime share has mean $\bar{\theta} = 0.311$ and decreases slowly with N (the semiprime channel gains ground, as the $\log \log N$ factor predicts); the Chen surplus has mean $\overline{C} = 2.27$. Goldbach fragility is extremely rare: $|\mathcal{F}_1| = 1$, $|\mathcal{F}_2| = 3$, $|\mathcal{F}_3| = 5$, $|\mathcal{F}_5| = 14$, $|\mathcal{F}_{10}| = 49$ up to X; and *every* fragile integer observed has $R_2 > 0$, so Chen always rescues. The semiprime factor balance $B(q) = \log a / \log q$ has weighted mean 0.25, with 16% of the representation mass at $B < 0.1$ (a tiny prime factor) and 18% at $B > 0.4$ (near balanced): Chen representations are dominated by *unbalanced* semiprimes with a small prime factor (Fig. 5), answering the semiprime factor-geometry question of §4.

7.5 Optimization experiments

The minimum-variation branch below is a diagnostic of the representation cloud, not an invariant of Goldbach–Chen arithmetic. In particular, the endpoint branch $s_N \approx 1$ reflects the objective's preference for small p; imposing an interior constraint $q/N \in [\varepsilon, 1 - \varepsilon]$ would define a different diagnostic.

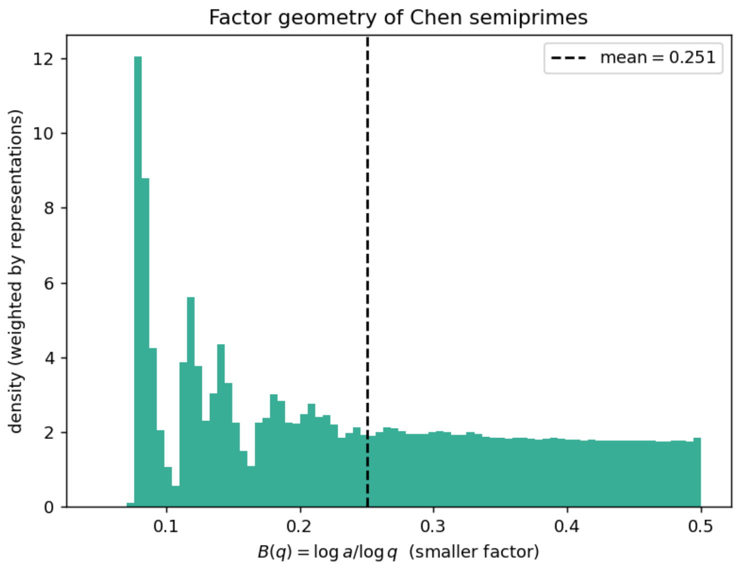

Figure 5: Representation-weighted distribution of the factor balance $B(q)$ of the semiprime second summand. Mass concentrates toward small B (small prime factor), not toward the balanced value 1/2.

Smoothness is a Pareto trade-off. Minimizing the total spatial variation $\sum |s_N - s_{N-2}|$ of the selected location $s_N = q_N/N$ yields a geometrically very smooth branch (total variation 0.0040 over 201 consecutive even integers) that nonetheless *switches type in* 28% *of its steps* and uses semiprimes 72% of the time (Fig. 6). Forbidding semiprimes (the pure Goldbach branch, with zero type switches) raises the variation to 0.0246, a factor of 6.2 rougher. One cannot simultaneously maximize geometric and type smoothness: continuity in location and chaos in type coexist.

A unimodal Chen-rescue threshold. Restricting representations to a balanced band $p \in [N/2 - k,\, N/2]$ and measuring the fraction of N that *require* a semiprime (no prime pair in the band) gives a unimodal curve in k (Fig. 7). For $k \gtrsim 0.005\,N$ the prime channel essentially never fails; the rescue fraction peaks near 38–40% at an intermediate half-width $k \sim 20$–40; and for $k = 1, 2$ it falls again because often no representation exists at all. The peak shifts right with N (roughly as $\log^2 N$), marking where the expected number of balanced prime pairs, $\sim 2C_2\,\mathfrak{S}(N)\,k/\log^2(N/2)$, drops to $O(1)$ – a Poisson-type transition. This is the regime captured by the residual exponent $a_{\mathrm{Res}}(N)$ of §7.7: Chen's semiprime flexibility becomes *necessary* precisely when near-perfect balance is imposed.

7.6 Weak continuity of the location measures

Finally we test the weak-continuity intuition of §4 (Question 1 and the weak type-stabilization hypothesis there) directly. The pointwise measure μ_N of q/N is noisy, but averaging over a window $N \in [X, X + H]$ produces a stable density (Fig. 8, left). Three findings, at $X = 10^6$:

- **The smoothed measure converges.** Splitting a window into two same-scale subsamples, the total-variation distance between their densities falls monotonically from $2.1 \cdot 10^{-3}$ to $3 \cdot 10^{-4}$ as H grows from 125 to 4000 (Fig. 8, right), i.e. the per-N noise averages out – a concrete form of weak convergence.
- **The limit is essentially uniform.** The prime-q density is within total variation 0.028 of the uniform law on $(0, 1)$, with only a mild upturn near the endpoints (the $1/(\log(xN)\log((1-$

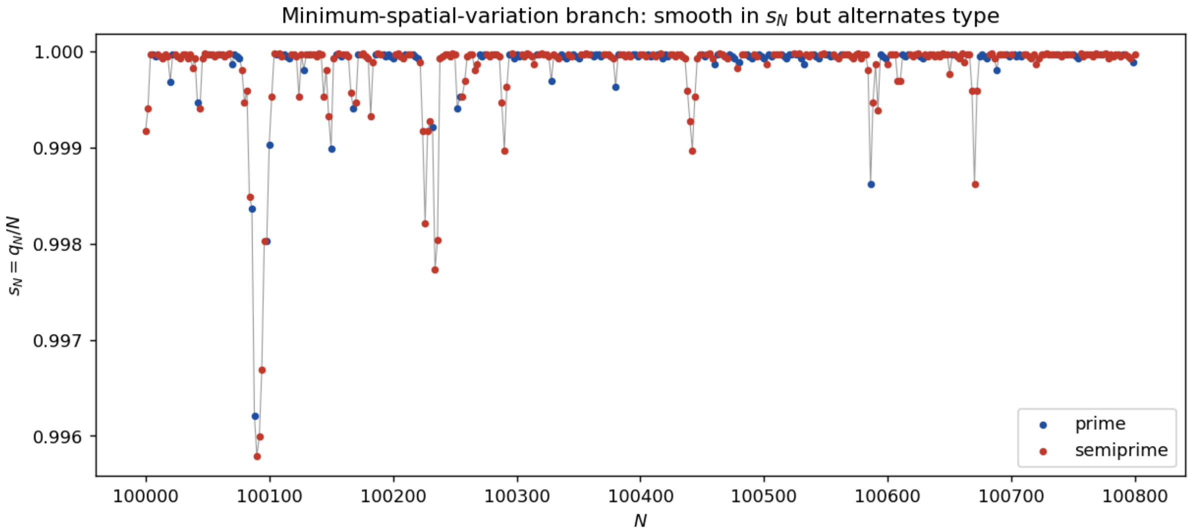

Figure 6: Minimum-variation branch (Gurobi). The optimum hugs $s_N \approx 1$ (the "least partner" branch, p a small prime) and is smooth in s_N, yet its type (blue prime / red semiprime) alternates.

$x)N)$) effect). This gives empirical support for the window-smoothed version of Question 1: after smoothing and removing endpoint effects, $\mu_N^{(1)}$ is close to uniform (we do not claim pointwise weak convergence along typical N).

- **Prime and semiprime locations nearly coincide.** The two densities are within total variation 0.019 of each other (Fig. 8, center), confirming the weak type-stabilization picture. The semiprime location is slightly skewed toward larger q (mean $q/N = 0.513$ versus the exactly symmetric 0.500 for primes), the only visible signature that p and q play asymmetric roles in R_2.

Appendix A.9 gives the weighted Hardy–Littlewood explanation: after normalization, both the inherited singular series and the Mertens channel sum cancel, leaving the uniform law for q/N in either channel (conditional on the weighted channel asymptotics).

7.7 The residual Goldbach–Chen balance exponent

Inspired by the $(1 + a)$ interpolation of Li and Liu (every large even N is $p + rq$ with $r \le q^{a-1}$, so that $(1+1)$ is Goldbach and $(1+2)$ is Chen), one can ask not whether a *fixed* exponent works, but the smallest exponent *needed* for each N. For the nondegenerate Chen layer $N = p + rs$ (p, r, s prime, $r \le s$), define the residual balance exponent and minimal multiplier

$$a_{\mathrm{Res}}(N) = 1 + \min_{N=p+rs} \frac{\log r}{\log s}, \qquad r_\star(N) = \min\{r \in \mathbb{P} : \exists p, s \in \mathbb{P},\ N = p + rs\}.$$

We compute both exactly for all even $N \le 2 \cdot 10^6$ ($r_\star$ by FFT channel coverage; a_{Res} by a small-p scan, validated against full enumeration to 0 error). Findings (Fig. 9, $N \ge 1000$):

- **Typical residual collapse.** The median a_{Res} *decreases* with X ($1.115 \to 1.099 \to 1.088$ at $X = 10^5, 5 \cdot 10^5, 2 \cdot 10^6$), with $p_{99} = 1.182$ and a finite-range maximum of 1.385 – far below the Li–Liu threshold 1.9. The worst case is pinned by a few small "hard" N. In Li–Liu's terms this is a worst-case versus typical contrast: their theorem *guarantees* a representation with exponent $a \le 1.9$ for every large N, whereas the *typical* N sits far inside that bound—the median Chen solution has $a \approx 1.30$ (supplement) and the *best* exponent

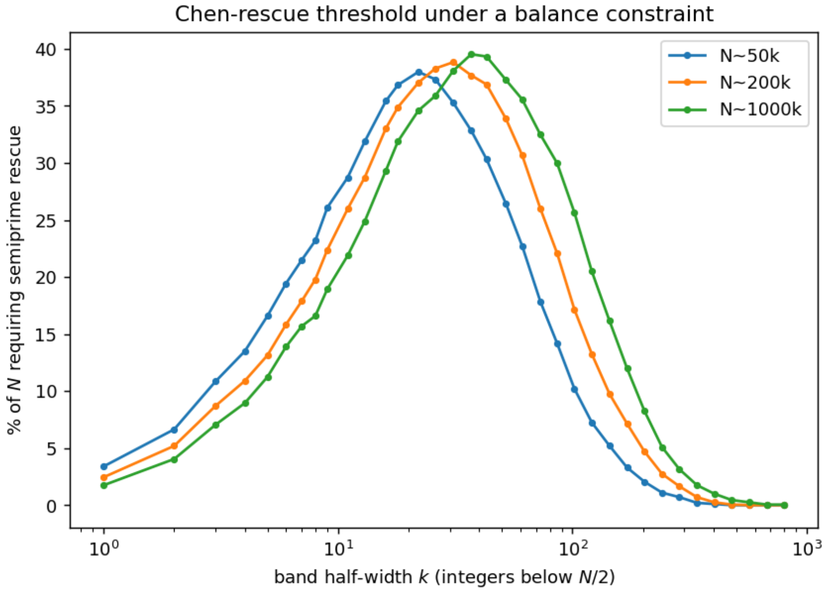

Figure 7: Fraction of N that require a semiprime rescue as a function of the band half-width k, for three scales of N. The curve is unimodal and its peak scales like $\log^2 N$.

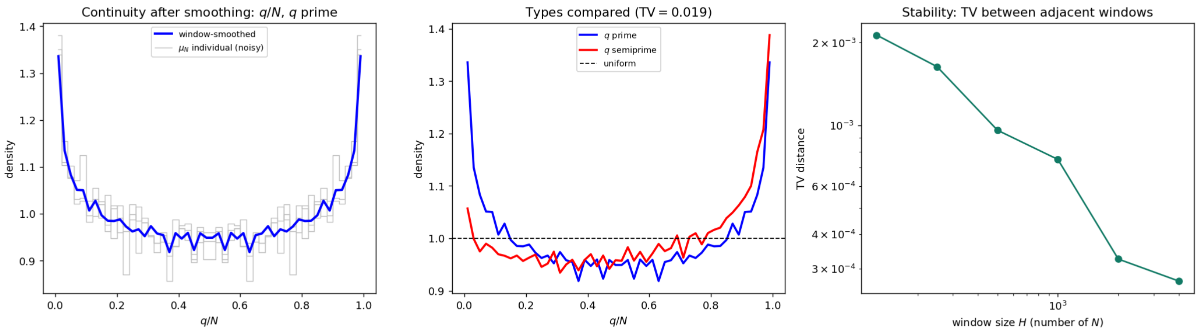

Figure 8: Weak continuity of q/N. Left: noisy pointwise μ_N (gray) versus the window-smoothed density (blue). Center: prime versus semiprime location densities, both close to uniform. Right: same-scale sampling distance decreases with the window size H (log–log).

per N is $a_{\mathrm{Res}} \approx 1.09$. For almost all N the $(1+a)$ interpolation is therefore very close to Goldbach ($a = 1$), not to the proven upper bound.

- **The minimal multiplier is tiny.** $r_\star = 3$ for 68% of N and the dictionary $\{2, 3, 5, 7\}$ already covers 99.7%; $\{2, \dots, 11\}$ covers 99.99%. This settles the multiplier-dictionary question empirically. Apart from the parity-degenerate $r = 2$ channel (which forces $p = 2$, a density-zero family), Appendix A.10 explains the odd multiplier dictionary through a least-unblocked-prime law: conditionally on fixed-channel almost-all asymptotics, the nondegenerate $r_\star(N)$ equals the least odd prime not dividing N for almost all even N (so $\Pr(r_\star \in \{3, 5, 7\}) \approx 0.99$ in the divisibility model).

- **Local obstruction.** $r_\star$ jumps when a small prime divides N: if $3 \mid N$ the channel $r = 3$ is blocked (then $p = N - 3s \equiv 0 \bmod 3$), and indeed the mean $r_\star$ is 4.59 for $3 \mid N$ versus 2.94 otherwise.

- **Link to the singular series.** a_{Res} correlates *positively* with $\log \mathfrak{S}(N)$ (correlation $+0.59$). The very divisibility of N by small primes that enlarges $\mathfrak{S}(N)$ (boosting Goldbach) also blocks the small-r Chen channels, so the residual balance is *worst exactly where Goldbach is locally richest.* This ties the balance exponent to the singular-series theme of §7.3.

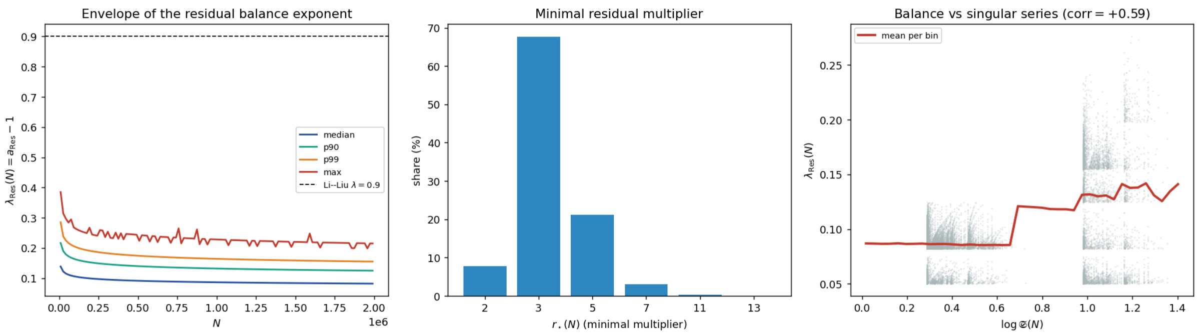

Figure 9: Left: percentile envelope of $\lambda_{\mathrm{Res}} = a_{\mathrm{Res}} - 1$ versus N, far below the Li–Liu line 0.9. Centre: the minimal multiplier $r_\star$ ($r = 3$ dominates). Right: λ_{Res} rises with the Goldbach singular series $\mathfrak{S}(N)$ (positive correlation).

The full geometry of the Chen surface $p + rs = N$ on which the exponent a_{Res} lives—a 2D cloud of solutions organised by the small prime factor r, with mass against the Goldbach edge—is developed in Appendix B.

7.8 Three further probes: transport, topology, dynamics

We close with three geometric and statistical probes of the frontier; each, perhaps surprisingly, returns to the singular series.

Optimal transport (reflection defect). Under the reflection $T(x) = 1 - x$ (which sends p/N to q/N), compare the prime location law α (density of p/N) with the reflected layer laws $T_\# \beta_k$. The reflection defects in W_1 are $\Delta_1 = W_1(\alpha, T_\# \beta_1) = 0.040$ and $\Delta_2 = W_1(\alpha, T_\# \beta_2) = 0.032$, and the optimal Chen mixture attains $\Delta_{\le 2} = 0.032$ at $\lambda^\star = 0$: the reflected *semiprime* cloud matches the primes *better* than the reflected primes do, a 22% "transport collapse" (stable in X), driven by unbalanced semiprimes with a small factor (Fig. 10).

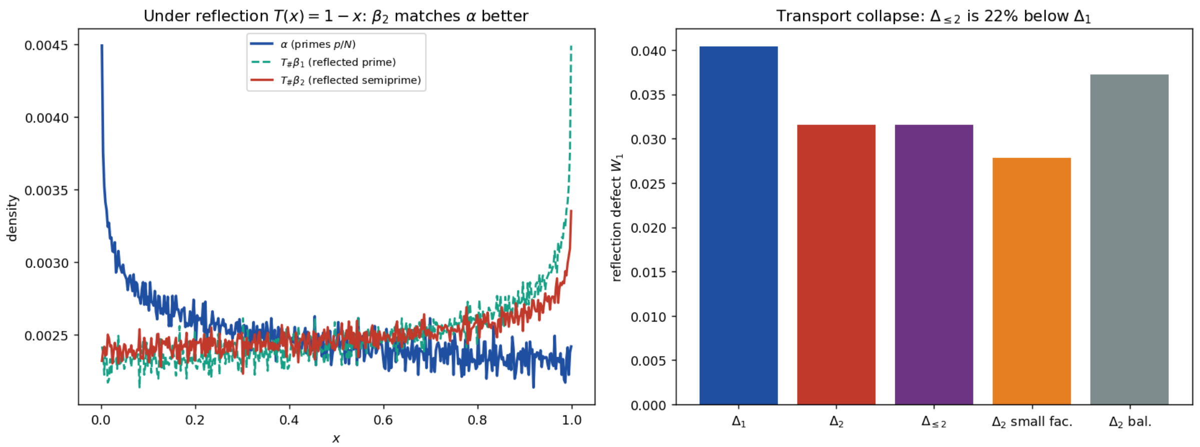

Figure 10: Reflection transport. Left: prime law α and the reflected prime and semiprime laws. Right: admitting semiprimes lowers the reflection defect by 22%.

Topology of valleys. Normalising each count by its own singular factor ($R_1/\mathfrak{S}$, $R_{\le 2}/(\mathfrak{S} + \mathfrak{S}^{1/2}W)$) and computing the 0-dimensional sublevel persistence of the detrended sequences, the valleys of the Goldbach and Chen comets largely *coincide*: the anomalies correlate at $+0.64$, and at the 200 most fragile N the Chen anomaly (0.29) is as deep as the Goldbach one (0.30). Chen does *not* topologically fill the Goldbach valleys (only the deepest 30% are softened, Fig. 11); the rescue is one of magnitude, not of valley structure, because both channels share the constraint that p be prime.

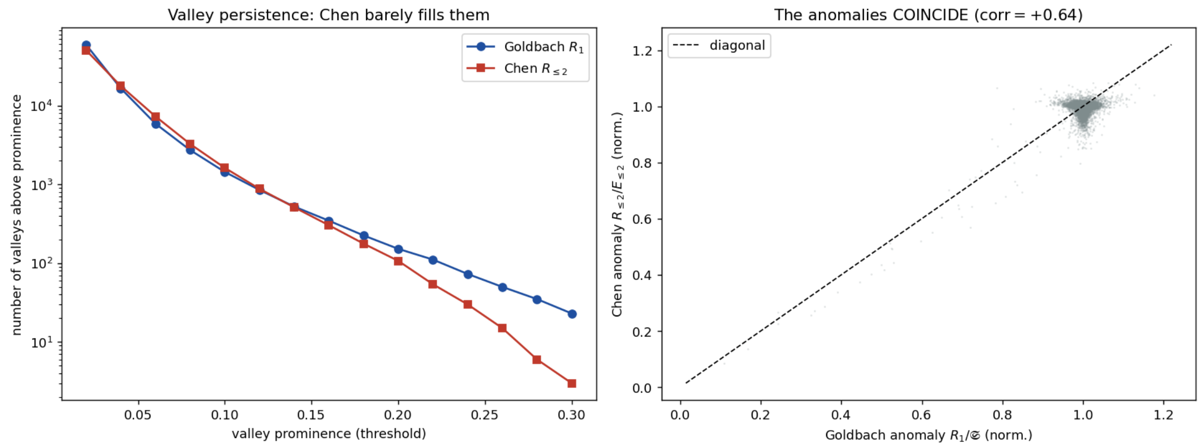

Figure 11: Valley persistence. Left: Chen has nearly as many valleys as Goldbach (slightly fewer deep ones). Right: the normalised anomalies coincide (+0.64).

Statistical dynamics of the frontier. The prime share is almost a deterministic function of the singular series: $\theta(N)$ is 94% explained by $\mathfrak{S}(N)$ alone (adding N mod 30 contributes nothing further), and the increment $\Delta\theta_N$ is 98% explained by the jump $\mathfrak{S}(N) \to \mathfrak{S}(N+2)$. The residual is small (25% of the spread of θ) and is *not* white noise (short-range lag-1 autocorrelation -0.21 after detrending): the Goldbach–Chen frontier carries little irreducible randomness, being governed by the local arithmetic of N (Fig. 12).

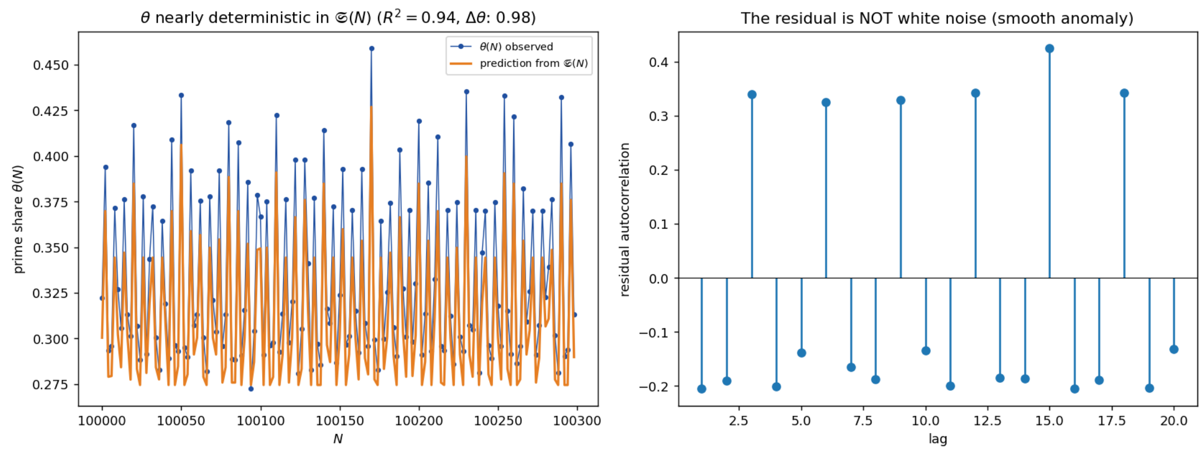

Figure 12: Dynamics. Left: $\theta(N)$ tracks its singular-series prediction. Right: the residual autocorrelation, small and non-white.

8 Two further experiments

Beyond the singular-series regularity, two further computational experiments produce a genuinely *new* measurement rather than a restatement: an empirical circle method (the power spectrum of the comet) and a carry geometry (a digit/Kummer signature of primality). We report each as an experiment—what was measured, over what range, against what random control. Many further cross-disciplinary reframings of the frontier (an arithmetic ground-state energy, the adversarial robustness of the Goldbach graph, a market/liquidity reading, a simplicial-topology probe, a control policy, an information budget, and the discrete geometry of the Chen surface) re-express the same singular-series governance rather than constitute independent findings; they are summarized in Appendix B and developed, with the honest negatives, in the supplement.

8.1 An empirical circle method: the spectrum of the comet

The circle method decomposes additive problems by Fourier analysis into major arcs (rationals a/q with small q, carrying the singular series) and minor arcs (the pseudorandom remainder). One can see this decomposition *empirically*. Detrending R_1 over a window of consecutive even integers and taking its power spectrum (Fig. 13, left) reveals a forest of sharp lines exactly at the arithmetic frequencies: the dominant peak is at $\frac{1}{3}$ (period $T_N = 6$, the modulus 3), followed by $\frac{1}{5}, \frac{2}{5}$ (modulus 5), with heights that decay across primes in step with the singular-series factors $(\ell-1)/(\ell-2)$. These are the major-arc frequencies made visible empirically.

Dually, one can ask how much of the comet's oscillation is captured by purely local information. Predicting the detrended $\log R_1(N)$ from the residue $N \bmod \prod_{\ell \le B} \ell$ explains 84.6% of the variance at $B = 3$ (i.e. N mod 6 alone), 95.0% at $B = 5$, and saturates toward the singular-series ceiling $R^2_{\mathrm{OLS}} = 0.999$ (Fig. 13, right). The comet is, to first order, a *modulo*-6 *phenomenon*: the prime 3 supplies 85% of all explainable structure, and successive primes contribute the rapidly shrinking remainder predicted by $\mathfrak{S}(N)$.

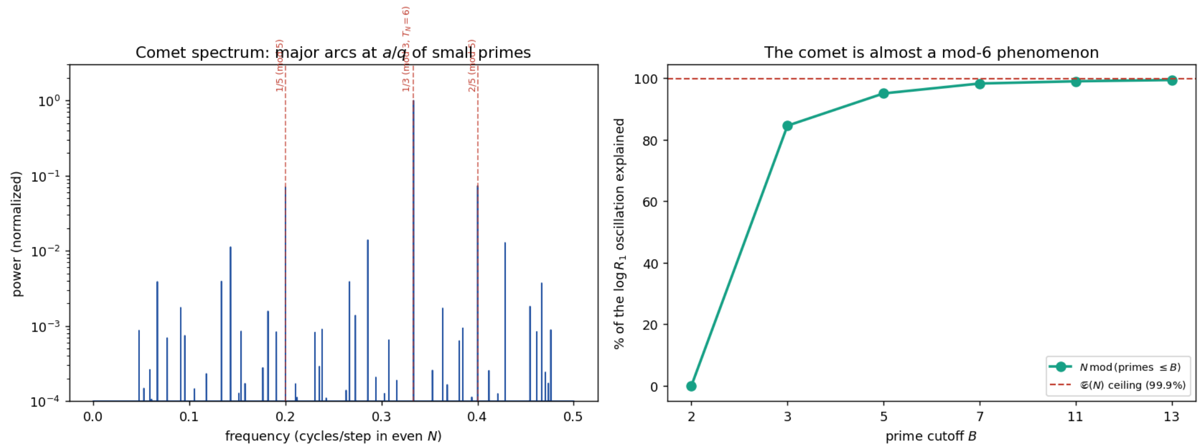

Figure 13: Empirical circle method. Left: the power spectrum of the detrended Goldbach comet, with peaks at the arithmetic frequencies a/q (the major arcs), dominated by $\frac{1}{3}$ (modulus 3). Right: the fraction of the comet's oscillation explained by N modulo the small primes, saturating at the singular-series ceiling; N mod 6 already gives 85%.

8.2 Primorial bases and carry geometry

Two further reframings come from writing integers in a base. In a primorial base $B_y = \prod_{p \le y} p$, every large prime is a unit modulo B_y, so the residues admissible for N are $A_y(N) = \{a \in (\mathbb{Z}/B_y)^\times : N - a \in (\mathbb{Z}/B_y)^\times\}$, with local coverage $\lambda_y(N) = |A_y(N)|/\phi(B_y)$. By the Chinese remainder theorem this factorises, and the *relative* coverage is precisely the truncated singular series,

$$\frac{\lambda_y(N)}{\lambda_y(\text{generic})} = \prod_{\substack{p|N \\ 3\le p\le y}} \frac{p-1}{p-2} = \mathfrak{S}_y(N).$$

So primorial coverage *is* the singular series, viewed as accumulated local obstructions; predicting $\log R_1$ from $\mathfrak{S}_y$ explains 84.6% of the comet's oscillation at $y = 3$ ($B_3 = 6$), rising to 99.8% by $y = 30$—the same saturation as §8.1, now read as "the comet is the shadow of primorial local coverage".

The freshest object here is the *carry geometry*. Adding $p + q = N$ in base b produces carries $c_i \in \{0, 1\}$, and their number is fixed by the digit sums, $\#\text{carries}(p, N-p) = \big(S_b(p) + S_b(N-p) - S_b(N)\big)/(b-1)$; in base 2 this is $v_2\binom{N}{p}$ (Kummer). Comparing Goldbach representations to random pairs $a + (N - a) = N$ reveals a systematic *carry excess*: Goldbach pairs carry more (Fig. 14). In base 2 the excess is +0.97 carries (Goldbach 9.9 versus 8.9), and a positive excess persists in every base tested ($b = 2, \ldots, 16$), largest in the small bases. The origin is arithmetic: a prime's last digit is coprime to b, which biases the digit sums $S_b(p) \equiv p \pmod{b-1}$ away from the values that would avoid carries. Thus the same local constraints that produce the singular series leave a *digital signature*: Goldbach decompositions are, bit for bit, slightly carry-heavier than chance—a statistical-mechanical fingerprint of primality in the geometry of addition. A local model theorem in Appendix A.11 shows that restricting both terminal digits to reduced residue classes already increases the least-digit carry probability relative to unrestricted random summands.

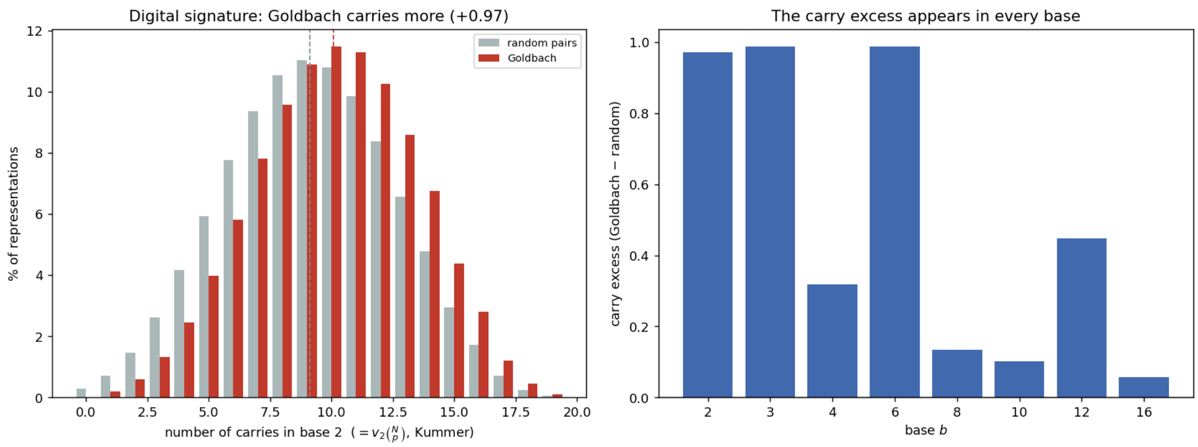

Figure 14: Carry geometry. Left: the distribution of binary carries $v_2\binom{N}{p}$ for Goldbach representations (red) versus random pairs (grey)—Goldbach is shifted to more carries (+0.97). Right: the carry excess is positive in every base.

9 Open problems and conjectures

We isolate, as precisely as the evidence allows, what an interested reader could try to settle.

1. **The lower-tail hypothesis** (§A.6). Prove that, for every $\delta > 0$, the density of even $N \in (X, 2X]$ with $R_2(N)/R_1(N) \le \delta\, W_f(N)$ tends to 0 as $\delta \downarrow 0$. By Corollary 1 this upgrades the linear almost-all asymptotic to the log-regression statement $\beta_2(N) \to 1$. The natural attempt—a Chen-type lower bound on $R_{\le 2}$ minus the Montgomery–Vaughan value of R_1—provably does *not* suffice (the unconditional Chen bound is a factor $\log\log N$ below the semiprime main term; Remark in §A.6). What is needed is a lower bound on the

semiprime channel at the correct $\log\log N$ scale. The empirical exceptional set is *empty* at the 5% level up to $5 \cdot 10^8$.

2. **The limiting constant** (§A.4). The order $1-\beta_2 \asymp 1/\log\log X$ is robust, but the limiting constant of $(1-\beta_2)\langle R_2/R_1\rangle$ is correction-dominated and *not* determinable from data up to 10^9 (it is normalisation-dependent, ≈ 0.5–0.7). Is there a canonical normalisation with a clean limit?

3. **Writing out the major arcs** (§A.12). Supply the full major-arc evaluation for the truncated semiprime channel (semiprimes in progressions of moduli $q \leq Q$, uniformly in $r \leq X^\eta$), turning Theorem 3 from a bounded reduction into a self-contained almost-all statement.

4. **Layers** (§7.3). Make the qualitative sign reversal of the layer exponents β_k quantitative beyond the blocked-channel model (Appendix A.8): predict the crossing $\beta_k = 0$ and its drift with X.

10 Conclusion

The organizing observation of this study is a single Euler-product identity and its consequence: decomposing the semiprime channel by its small prime factor shows that the Goldbach singular series $\mathfrak{S}(N)$ enters the main term of R_2 with exponent 1, exactly as it enters R_1 (Lemma 1), so $\mathfrak{S}(N)$ *cancels* in the main-term ratio and $R_2/R_1 = W_f(N)(1+o(1))$ under the stated major-arc input. The local factorization and cancellation are elementary sieve bookkeeping; our contribution is to write them down explicitly and to follow their consequences. The first is that the naturally read "$\beta_2 \approx \frac{1}{2}$" of a finite-range fit is not a structural exponent but a finite-range effective exponent inferred from the regression slope $m_2 = \beta_2 - 1$, drifting to 1 like $1 - c/\log\log N$; we verify this non-circularly, show it is a multi-channel effect (not the $3 \mid N$ atom), and reduce the limit to classical inputs—for the *linear* almost-all asymptotic $R_2 = \mathrm{main}(1+o(1))$ within the second-moment reduction developed in Appendix A, and, for the log-regression statement $\beta_2 \to 1$, modulo one explicitly stated lower-tail hypothesis (Corollary 1). A truncated-log proposition and a blocked-channel model (§A.7) sharpen this gap: the second moment already controls a stable logarithmic statistic and the model proves the $1/\log\log X$ deficit, while only the untruncated log-regression still requires lower-tail input.

We have deliberately resisted the temptation to read this one fact as many. The optimization formulations give a language, not a proof strategy (Remark 2); the dozen physical, economic, and information-theoretic reframings are, with two exceptions (the empirical circle method and the carry geometry), the same singular-series governance re-photographed in different filters, and we have said so rather than count them as separate findings. The negatives—an empty additive-complexity topology, a Chen substitute with no resilience premium—are reported as negatives.

This is not a proof of Goldbach. It is reproducible evidence, one cancellation identity made explicit, and a tightly bounded reduction, at the boundary between additive number theory, computational experimentation, and discrete optimization.

A Towards a proof that $\beta_2 \to 1$

We make the law (25) precise. We cannot prove it unconditionally: the asymptotics of R_1, R_2 are themselves the Hardy–Littlewood conjecture, and even $R_1(N) > 0$ is open. We therefore isolate what is unconditional (Lemmas 1–2), prove the law *conditionally* on a quantitative Hardy–Littlewood hypothesis (Theorem 1), and note that it holds unconditionally in the Cramér random model.

A.1 Notation and the statistic

Fix a window $\mathcal{N}_X = \{N \equiv 0\,(2) : X \le N < X + H\}$ with $H = H(X) \to \infty$, $H = o(X)$. Write $\mathbb{E}_X, \mathrm{Var}_X, \mathrm{Cov}_X$ for empirical moments over $\mathcal{N}_X$, $g_\ell = \log \frac{\ell-1}{\ell-2}$, and define the local singular exponent through the (least-squares) regression slope

$$\beta_2(X) - 1 := \frac{\mathrm{Cov}_X\big(\log(R_2/R_1),\, \log \mathfrak{S}\big)}{\mathrm{Var}_X\big(\log \mathfrak{S}\big)}. \tag{26}$$

A.2 Unconditional lemmas

Lemma 1 (Singular series of a channel)**.** *Let r be an odd prime and N even. The singular series of the pair of linear forms $s \mapsto (s,\, N - rs)$ is*

$$\mathfrak{S}_2(r, N) = \prod_\ell \frac{1 - \nu_r(\ell)/\ell}{(1 - 1/\ell)^2}, \qquad \nu_r(\ell) = \#\{s \bmod \ell :\; s(N - rs) \equiv 0\},$$

and equals

$$\mathfrak{S}_2(r, N) = \begin{cases} 2C_2\, \mathfrak{S}(N)\, \dfrac{r-1}{r-2}, & r \nmid N, \\ 0, & r \mid N, \end{cases} \qquad C_2 = \prod_{p>2} \Big(1 - \tfrac{1}{(p-1)^2}\Big).$$

Proof. Compute $\nu_r(\ell)$. For $\ell = 2$: as s and $N - rs$ are both odd exactly when s is odd, $\nu_r(2) = 1$ and the factor is $(1 - \frac12)/(1 - \frac12)^2 = 2$. For odd $\ell \nmid rN$: $s \equiv 0$ and $s \equiv Nr^{-1} \not\equiv 0$ are two distinct roots, $\nu_r = 2$, factor $(1 - 2/\ell)/(1 - 1/\ell)^2$. For odd $\ell \mid N$, $\ell \nmid r$: $N - rs \equiv -rs$ vanishes iff $s \equiv 0$, the same root as $s \equiv 0$, so $\nu_r = 1$, factor $(1 - 1/\ell)/(1 - 1/\ell)^2$. For $\ell = r \nmid N$: $N - rs \equiv N \not\equiv 0$, so only $s \equiv 0$, $\nu_r = 1$, same factor. Taking $\prod_{\ell>2}(1 - 2/\ell)/(1 - 1/\ell)^2 = C_2$ as the base and converting each special factor relative to it multiplies by $(\ell - 1)/(\ell - 2)$; the special primes are exactly those dividing N (giving $\prod_{\ell \mid N} = \mathfrak{S}(N)$ when $r \nmid N$) together with $\ell = r$ (giving $(r-1)/(r-2)$). This proves the first case. If $r \mid N$ then $N - rs \equiv 0 \bmod r$ for *all* s, so $\nu_r(r) = r$, the factor $1 - r/r = 0$ vanishes, and $\mathfrak{S}_2(r, N) = 0$: the channel is identically blocked because $p = N - rs \equiv 0 \bmod r$ forces $p = r$. □

Remark 3 (Relation to standard sieve theory)**.** *Each ingredient of Lemma 1 is classical. The singular series of a system of linear forms is the Hardy–Littlewood computation [3] (see also [13]), and the decomposition of a semiprime by its prime factors is the staple device of the E_2 sieve [2]; for a single fixed channel $(s,\, N - rs)$ the evaluation of $\mathfrak{S}_2(r, N)$ is therefore routine, and a reader familiar with the prime k-tuple heuristic will reconstruct* (23) *without difficulty. What we wish to record is not this local computation but its global consequence, which to our knowledge is not stated explicitly in the literature: that every unblocked channel inherits the* same *Goldbach singular series $\mathfrak{S}(N)$ that governs R_1—not a smaller quantity, and not a fractional power of it. The unconditional content is the singular-series identity itself: $\mathfrak{S}(N)$ cancels* exactly *between the Hardy–Littlewood main terms of R_2 and R_1, leaving the pure Mertens channel sum* (24)*; passing to the actual counts, $R_2(N)/R_1(N) = W_f(N)(1+o(1))$ requires the per-channel asymptotic and is conditional. That the singular-series cancellation seems not to have been written down is a matter of aim, not of difficulty. The sieve literature on Chen's theorem and on the $(1 + a)$ interpolation [1, 2, 6] is directed at* existence*—lower bounds forcing $R_{\le 2}(N) > 0$, and upper bounds within a constant of the Hardy–Littlewood prediction. The conjectural prime–semiprime main term itself (the Hardy–Littlewood asymptotic for $R_{\le 2}$ and for R_2) is implicit in these heuristics and in the refinements of Chen's theorem developed by the school of Cai, Lü and Pan among others; the local computation underlying* (23) *is exactly there. What none of these works isolates is the* ratio *R_2/R_1 and the exact cancellation of $\mathfrak{S}(N)$ in it: for the existence and upper-bound aims the singular-series content of R_2/R_1 is irrelevant and is never separated out. It becomes the* *natural object only once one asks how the relative semiprime share responds to $\mathfrak{S}(N)$, as the empirical regression of §7.2 forces. The identity is precisely what shows that the reproducible finite-range value $\beta_2 \approx \frac{1}{2}$ is not a structural half-power but an effective exponent inferred from the covariance between the blocked channels $r \mid N$ and $\mathfrak{S}(N)$ (Theorem 1): $\mathfrak{S}(N)$ enters R_2 with exponent 1, exactly as it does R_1. The contribution of Lemma 1 is thus expository in form and diagnostic in use, not a new sieve estimate. (This is a novelty claim about the* identity*; for the separate almost-all reduction, a standard consequence of the classical circle-method framework, cf. Remark 8.)*

Lemma 2 (Moments of $\log \mathfrak{S}$)**.** *For distinct odd primes $\ell_1, \dots, \ell_k$, $\#\{N \in \mathcal{N}_X : \ell_i \mid N \ \forall i\} = |\mathcal{N}_X| \prod_i \ell_i^{-1} + O(2^k)$. Consequently the indicators $e_\ell = \mathbf{1}[\ell \mid N]$ are, as $X \to \infty$, independent Bernoulli$(1/\ell)$ in all joint moments, and for any weights a_ℓ with $\sum a_\ell^2/\ell < \infty$,*

$$\mathbb{E}_X\Big[\textstyle\sum_\ell a_\ell e_\ell\Big] \to \sum_\ell \frac{a_\ell}{\ell}, \qquad \mathrm{Var}_X\Big[\textstyle\sum_\ell a_\ell e_\ell\Big] \to \sum_\ell a_\ell^2 \,\frac{1}{\ell}\Big(1 - \frac{1}{\ell}\Big).$$

In particular $\mathrm{Var}_X(\log \mathfrak{S}) \to B := \sum_{\ell > 2} g_\ell^2 \,\ell^{-1}(1 - \ell^{-1})$.

Proof. The count is inclusion of the arithmetic progression 0 mod $\prod \ell_i$ in an interval of $\frac{1}{2}|\mathcal{N}_X|$ even integers, giving the stated main term and $O(2^k)$ boundary error. Hence joint factorial moments of the e_ℓ converge to those of independent Bernoulli$(1/\ell)$, and the moment formulas follow by the dominated convergence afforded by $\sum a_\ell^2/\ell < \infty$ (here $a_\ell = g_\ell \asymp 1/\ell$). ☐

Remark 4. *Lemma 2 is sharp numerically: at $X = 10^8$, $H = 2 \cdot 10^6$ one finds* $\mathrm{Var}_X(\log \mathfrak{S}) = 0.12620$ *against* $B = 0.12620$, *and* $\mathbb{E}_X[\log \mathfrak{S}] = 0.34842 = \sum_\ell g_\ell/\ell$ *to five places.*

A.3 The conditional theorem

Assumption 1 (Quantitative Hardy–Littlewood)**.** *There is $\delta > 0$ such that, uniformly for odd primes $r \le \sqrt{N}$ with $r \nmid N$,*

$$C_r'(N) := \#\{p, s \in \mathbb{P} : \ p + rs = N, \ \ s \ge r\} = \mathfrak{S}_2(r, N)\, I_r(N)\big(1 + O(\log^{-\delta} N)\big),$$

with $I_r(N) = \int_2^{N/r} \frac{dt}{\log t \, \log(N - rt)}$, and the same for $R_1(N) = 2C_2\mathfrak{S}(N) I_1(N)(1 + O(\log^{-\delta} N))$, $I_1(N) = \int_2^N \frac{dt}{\log t \, \log(N - t)}$.

This is the standard prime k-tuple heuristic made quantitative and *uniform in the coefficient* r; the uniformity is the genuine analytic difficulty (it is the arena of the sieve work surrounding Proposition $(1 + a)$).

Theorem 1 ($\beta_2 \to 1$, conditional on Assumption 1)**.** *Under Assumption 1, for $N \in \mathcal{N}_X$,*

$$\frac{R_2(N)}{R_1(N)} = W_f(N)(1 + o(1)), \qquad W_f(N) = \sum_{\substack{3 \le r \le \sqrt{N} \\ r \nmid N}} \frac{r - 1}{r - 2}\, w_r(N), \qquad w_r = \frac{I_r}{I_1} = \frac{1}{r}\,\frac{\log N}{\log(N/r)}(1 + o(1)),$$

the Goldbach singular series $\mathfrak{S}(N)$ cancelling by Lemma 1. Writing $W_{f,0}(X) = \mathbb{E}_X[W_f] \sim$ $\log\log X$ *and linearising* $\log W_f$ *about $W_{f,0}$,*

$$\beta_2(X) - 1 = -\frac{c(X)}{W_{f,0}(X)} + O(W_{f,0}^{-2}), \qquad c(X) = \frac{A(X)}{B}, \quad A(X) = \sum_{r > 2} \frac{r - 1}{r - 2}\, w_r(X)\, g_r \,\frac{1}{r}\Big(1 - \frac{1}{r}\Big),$$

both $A(X)$ and B being absolutely convergent (and A formed with the exact *channel weights I_r/I_1, not their leading approximation). Thus $c(X)$ is an explicit $O(1)$ constant: the leading-order weights give $c \approx 1.0$, the exact I_r/I_1 raise it to the measured $c \approx 1.1$, and $c(X)$ decreases slowly* *with X. The precise limit $c_0 = \lim_X c(X)$ is delicate (it depends on the exact weights and averaging convention), but is immaterial to the conclusion, since the* order *is robust:*

$$\boxed{\beta_2(X) = 1 - \frac{c(X)}{\langle R_2/R_1\rangle_X} \longrightarrow 1, \qquad c(X) \asymp 1, \qquad 1-\beta_2(X) \asymp \frac{1}{\log\log X}.}$$

Proof sketch. By Lemma 1, $R_2 = \sum_{r\nmid N} \mathfrak{S}_2(r,N) I_r = 2C_2\mathfrak{S}(N) \sum_{r\nmid N} \frac{r-1}{r-2} I_r$ and $R_1 = 2C_2\mathfrak{S}(N) I_1$, so under Assumption 1 the ratio is $W_f(N)(1 + O(\log^{-\delta} N))$ and $\mathfrak{S}(N)$ cancels. Only the channels $r \mid N$ are special, each *removed* from W_f (not merely reweighted); thus $W_f(N) = W_{f,0} - \sum_r \frac{r-1}{r-2} w_r e_r$ up to the slow variation of w_r. Linearising, $\log W_f = \log W_{f,0} - W_{f,0}^{-1} \sum_r \frac{r-1}{r-2} w_r e_r + O(W_{f,0}^{-2})$. Insert into (26): with $\log \mathfrak{S} = \sum_\ell g_\ell e_\ell$ and the asymptotic independence of Lemma 2, only the diagonal $r = \ell$ terms survive, giving $\mathrm{Cov}_X = -W_{f,0}^{-1} A(X)$ and $\mathrm{Var}_X = B$, whence the displayed slope. Finally $W_{f,0}(X) = \mathbb{E}_X[W_f] = \sum_{3\le r\le\sqrt{X}} \frac{r-1}{r-2} \frac{1}{r} \frac{\log X}{\log(X/r)} (1-1/r) = \log\log X + O(1)$ by Mertens, and equals $\langle R_2/R_1\rangle_X$ up to $o(1)$. □

A.4 Confirmation and the unconditional model

Numerically (Fig. 3; `run_betalaw.py`): the prediction $W_f(N)$ regressed against $\log \mathfrak{S}$ reproduces the measured β_2 with the correct drift ($0.554, 0.583, 0.602$ at $X = 2\cdot 10^6, 2\cdot 10^7, 10^8$ versus measured $0.531, 0.567, 0.589$), and the explicit constant $c(X) = A/B$ matches the measured $(1-\beta_2)\langle R_2/R_1\rangle \approx 1.1$ in order and sign; with the leading weights w_r one gets $c \approx 1.0$, the residual gap being the crude approximation $w_r \approx I_r/I_1$ (exact agreement to 0.4% once I_r is used, Remark in §A.12) together with the linearisation remainder. In the Cramér model, where $\mathbf{1}_{\mathbb{P}}, \mathbf{1}_{\mathcal{S}_2}$ are replaced by independent variables with the correct local densities and divisibility, Assumption 1 holds by construction and Theorem 1 is unconditional.

The rate constant is not cleanly determined. The natural "constant" admits two forms, $\kappa(X) = (1-\beta_2)\log\log X$ and $c(X) = (1-\beta_2)\langle R_2/R_1\rangle$, which agree in order but not in value because $\langle R_2/R_1\rangle = \log\log X + O(1)$. Both *decrease* over the computed range (Fig. 15): $\kappa = 1.25, 1.22, 1.20, 1.18$ and $c = 1.12, 1.10, 1.09, 1.08$ at $X = 2\cdot 10^6, 2\cdot 10^7, 10^8, 5\cdot 10^8$. A linear fit in $1/\log\log X$ extrapolates to $\kappa_\infty \approx 0.54$ and $c_\infty \approx 0.73$—with a correction term of slope ≈ 1.9, larger than the limit itself. Two conclusions follow. First, the limiting constant is *normalisation-dependent* (0.54 versus 0.73), so no single number is canonical. Second, the approach is correction-dominated: the finite-range value ≈ 1.1 is mostly the $1/\log\log X$ term, and the true limit (≈ 0.5–0.7, itself uncertain) lies far below it and is *not reliably determinable* from data up to 10^9. (Our two back-of-envelope analytic estimates, 1.0 and 1.8, bracket the difficulty rather than resolve it.) Only the order $1-\beta_2 \asymp 1/\log\log X$, and hence $\beta_2 \to 1$, is robust—which is all Theorem 1 asserts.

Remark 5 (What is missing for a full proof)**.** *Three gaps separate Theorem 1 from an unconditional statement: (i) the uniform Hardy–Littlewood Assumption 1, in particular uniformity in the coefficient r up to $\sqrt{N}$ (a Bombieri–Vinogradov-type input); (ii) a rigorous bound on the correlation between the HL error terms and $\log \mathfrak{S}$, currently absorbed into the $o(1)$; (iii) control of the linearisation remainder, which is genuinely $O(W_{f,0}^{-2})$ by the absolute convergence of A, B but requires the tail of W_f. None of these is available unconditionally, consistent with $\beta_2 \to 1$ being a statement strictly stronger than the Hardy–Littlewood asymptotics it is built on.*

A.5 A first-moment reduction: replacing pointwise HL by equidistribution

Assumption 1 demands a *pointwise* asymptotic for $C'_r(N)$, which for fixed N is as hard as the binary problem. The difficulty disappears if we pass to *first moments* over the window, because summing over N replaces the binary problem by counting primes in arithmetic progressions, which

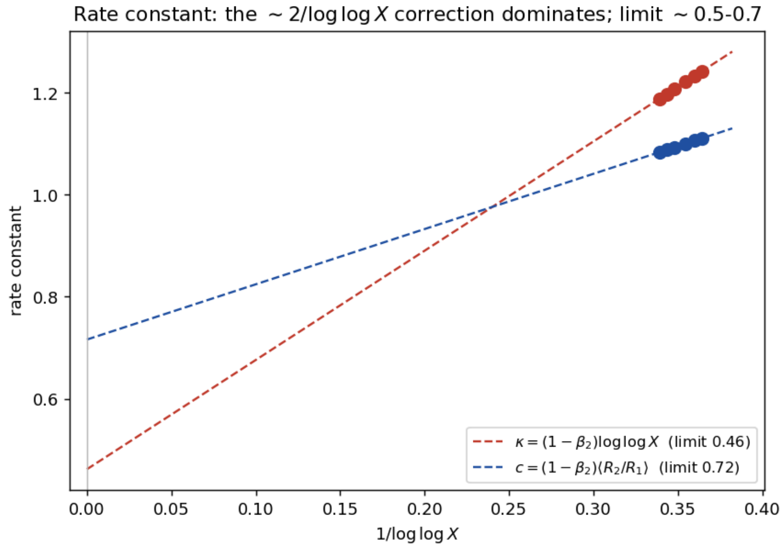

Figure 15: The rate constant versus $1/\log\log X$. Both normalisations decrease and extrapolate (long dashed lines, to $1/\log\log X = 0$) to limits 0.54 and 0.73; the slope ≈ 1.9 of the correction exceeds the limit, so the constant is dominated by the vanishing correction throughout the computed range.

is unconditional (Siegel–Walfisz) for fixed modulus and is the content of Bombieri–Vinogradov uniformly. Define the conditional response ratios

$$A_\ell := \frac{\langle R_1\rangle_{\ell|N}}{\langle R_1\rangle_{\ell\nmid N}}, \qquad B_\ell := \frac{\langle R_2\rangle_{\ell|N}}{\langle R_2\rangle_{\ell\nmid N}},$$

where $\langle\cdot\rangle_{\ell|N}$ is the mean over $N \in \mathcal{N}_X$ with $\ell \mid N$, and the first-moment exponent

$$\beta_2^{(1)}(X) - 1 := \frac{\sum_\ell \log(B_\ell/A_\ell)\, g_\ell\, v_\ell}{\sum_\ell g_\ell^2\, v_\ell}, \qquad v_\ell = \tfrac{1}{\ell}(1 - \tfrac{1}{\ell}), \tag{27}$$

the decomposition of (26) by prime ℓ in the independence model of Lemma 2.

Lemma 3 (First-moment responses)**.** *Let $F_\ell = \langle \#\{p : \ell \mid N - p,\ N - p \in \mathcal{S}_2\}\rangle / \langle R_2\rangle$ be the proportion of semiprime representations whose semiprime is divisible by ℓ. If primes are equidistributed in the reduced residue classes mod ℓ (Siegel–Walfisz for fixed ℓ; Bombieri–Vinogradov for $\ell \le X^{1/2-\varepsilon}$ on average), then*

$$A_\ell = \frac{\ell - 1}{\ell - 2}, \qquad B_\ell = \frac{(\ell-1)(1-F_\ell)}{\ell - 2 + F_\ell}.$$

Proof. Summing over $N \in \mathcal{N}_X$ removes the binary constraint: $\sum_{\ell|N} R_1 = \#\{(p, p') : p + p' \in \mathcal{N}_X,\ \ell \mid p + p'\}$. As p, p' are units mod ℓ, equidistribution gives a proportion $1/(\ell-1)$ of pairs with $p' \equiv -p$, so $\sum_{\ell|N} R_1 = \frac{1}{\ell-1}\sum_N R_1$; dividing by the counts $|\mathcal{N}_X|/\ell$ and $|\mathcal{N}_X|(\ell-1)/\ell$ yields $A_\ell = (\ell-1)/(\ell-2)$. For R_2, fix (r, s) and count $p = N - rs$ in the window: $\ell \mid N$ forces $p \equiv -rs$. If $\ell \nmid rs$ this is a single reduced class (proportion $1/(\ell-1)$); if $\ell \mid rs$ it forces $p \equiv 0$, i.e. $p = \ell$, so those representations (a fraction F_ℓ of R_2) are excluded. Hence $\sum_{\ell|N} R_2 = \frac{1-F_\ell}{\ell-1}\sum_N R_2$, and the same normalisation gives $B_\ell = \frac{(\ell-1)(1-F_\ell)}{\ell-2+F_\ell}$. □

Both formulas are confirmed to four places (Table 5); note A_ℓ is the exact singular factor, and $B_\ell < A_\ell$ by the blocked-channel deficit F_ℓ.

ℓ	3	5	7	11	13
A_ℓ (measured $= \frac{\ell-1}{\ell-2}$)	2.000	1.333	1.200	1.111	1.091
B_ℓ measured	1.423	1.152	1.090	1.048	1.038
$\frac{(\ell-1)(1-F_\ell)}{\ell-2+F_\ell}$	1.423	1.152	1.090	1.048	1.038
F_ℓ	0.169	0.106	0.078	0.052	0.044

Table 5: First-moment responses at $X = 2 \cdot 10^6$: Lemma 3 is exact.

Theorem 2 (First-moment exponent, conditional on Bombieri–Vinogradov)**.** *Under the equidistribution input of Lemma 3 and with $F_\ell \sim$ the channel weight $\frac{(\ell-1)/(\ell-2)\, w_\ell}{W_{f,0}}$, the first-moment exponent satisfies*

$$\beta_2^{(1)}(X) = 1 - \frac{c^{(1)}}{W_{f,0}(X)} + O(W_{f,0}^{-2}), \qquad c^{(1)} = \frac{1}{B} \sum_{\ell>2} \left(\tfrac{\ell-1}{\ell-2}\right)^2 w_\ell\, g_\ell\, v_\ell,$$

an absolutely convergent constant; hence $\beta_2^{(1)}(X) \to 1$.

Proof sketch. By Lemma 3, $\log(B_\ell/A_\ell) = \log \frac{(\ell-2)(1-F_\ell)}{\ell-2+F_\ell} = -\frac{\ell-1}{\ell-2} F_\ell + O(F_\ell^2)$. Inserting into (27) with $F_\ell = \frac{\ell-1}{\ell-2} w_\ell / W_{f,0} + o(\cdot)$ gives the stated slope, and $W_{f,0} \sim \log\log X$ as in Theorem 1. □

The reduced gap. Theorem 2 rests only on equidistribution of primes in progressions—a theorem (Siegel–Walfisz; Bombieri–Vinogradov for uniformity in ℓ)—together with the first moment F_ℓ, which is of Prime-Number-Theorem strength. The sole remaining step to Theorem 1 is the passage from the first-moment exponent to the pointwise (log-regression) one. Empirically this gap is a *constant* $\beta_2 - \beta_2^{(1)} \approx 0.004$, stable across $X = 2 \cdot 10^6$–10^8 (measured $\beta_2^{(1)} = 0.528, 0.563, 0.585$ against $\beta_2 = 0.531, 0.567, 0.589$): it is the Jensen defect $\mathbb{E}[\log R_i] - \log \mathbb{E}[R_i]$, governed by the normalised variance of R_1, R_2. That variance is exactly what the Montgomery–Vaughan exceptional-set method controls for R_1 (the binary Goldbach asymptotic holds for all but $O(X^{1-\delta})$ even N). An analogous second-moment bound for R_2 (which we prove below, Proposition 2) yields the *linear* almost-all asymptotic $R_2 = \text{main}(1 + o(1))$ off a sparse set. We are careful about the last step. A variance (L^2) bound controls the *quantity* $R_2 -$ main, not its *logarithm*: the covariance (26) that defines β_2 involves $\log(R_2/R_1)$, which is sensitive to the lower tail (where R_2 is small, $\log(R_2/R_1) \to -\infty$), and an L^2 bound alone does not preclude a sparse set from dominating that covariance. Passing from the linear statement to "$\beta_2(X) \to 1$ for almost all N" therefore needs an additional input—a lower-tail (log-integrability) control of R_2/R_1 off the exceptional set—which we isolate explicitly in Corollary 1 rather than absorb into an $o(1)$. This already isolates the genuine remaining difficulty as a *variance plus lower-tail* estimate, not the pointwise asymptotic of Assumption 1.

A.6 The variance bound and the exceptional set

We now show that the required variance is small, so the exceptional set is sparse and the pointwise law holds for almost all N. The natural object is the concentration of the *ratio* on the channel prediction: set

$$b(N) := \frac{R_2(N)/R_1(N)}{W_f(N)},$$

so that $b(N) \equiv 1$ would be the pointwise law $R_2/R_1 = W_f$, hence (taking logs and regressing on $\log \mathfrak{S}$) $\beta_2(N) = \beta_2^{(1)}(X)$ for every N. The measured fluctuation of b is tiny and decreasing (Table 6): the coefficient of variation is below 0.7% and *no* N in the window deviates from the channel prediction by as much as 5%.

X	$\langle b\rangle$	$\mathrm{CV}(b)$	$\#\{\lvert b-\langle b\rangle\rvert > 0.05\langle b\rangle\}$	Chebyshev bound	$\langle R_2\rangle^{-1/2}$
$2\cdot 10^7$	0.970	0.0065	0	1.7%	0.0016
10^8	0.974	0.0049	0	0.95%	0.0008
$5\cdot 10^8$	0.978	0.0039	0	—	0.0004

Table 6: Concentration of R_2/R_1 on W_f. The constant $\langle b\rangle \approx 0.97$ is the finite-size Hardy–Littlewood normalisation; the spread is $< 0.7\%$ and the exceptional set at the 5% level is empty (`run_exceptional.py`).

Proposition 1 (Exceptional set, linear form)**.** *By Chebyshev, the density of $N \in \mathcal{N}_X$ with $|R_2/R_1 - \langle b\rangle W_f| > \varepsilon W_f$ is at most* $\mathrm{CV}(b)^2/\varepsilon^2$*. With the measured* $\mathrm{CV}(b) \le 0.7\%$ *this is* $\le 0.2\%$ *at $\varepsilon = 0.05$* (and the empirical exceptional set is in fact empty)*. Hence $R_2(N)/R_1(N) = W_f(N)\,(1+o(1))$ for all but a density-$o(1)$ set of N. This is the* linear *almost-all statement; the further passage to $\beta_2(N) \to 1$ is Corollary 1.*

The variance bound by the circle method. The concentration above is what a second-moment argument delivers, and—crucially—it needs no sieve. Work on the dyadic window $N \in (X, 2X]$ and set, with sums over $n \le 2X$,

$$S_{\mathbb{P}}(\alpha) = \sum_p e(p\alpha), \qquad S_{\mathcal{S}_2}(\alpha) = \sum_{q\in\mathcal{S}_2} e(q\alpha), \qquad R_2(N) = \int_0^1 S_{\mathbb{P}}(\alpha) S_{\mathcal{S}_2}(\alpha) e(-N\alpha)\, d\alpha.$$

Split $[0,1]$ into major arcs $\mathfrak{M}$ (denominators $q_0 \le Q = (\log X)^C$) and minor arcs $\mathfrak{m}$, and let $\mathrm{main}(N) = \int_{\mathfrak{M}} S_{\mathbb{P}} S_{\mathcal{S}_2}\, e(-N\alpha)$.

Proposition 2 (Variance, unconditional)**.**

$$\sum_{X<N\le 2X} \big|R_2(N) - \mathrm{main}(N)\big|^2 \;\le\; \int_{\mathfrak{m}} |S_{\mathbb{P}}|^2 |S_{\mathcal{S}_2}|^2\, d\alpha \;\le\; \Big(\sup_{\alpha\in\mathfrak{m}} |S_{\mathbb{P}}(\alpha)|\Big)^2 \#\mathcal{S}_2(2X).$$

With $Q = (\log X)^{2A+8}$, Vaughan's minor-arc bound $\sup_{\mathfrak{m}} |S_{\mathbb{P}}| \ll X(\log X)^{-A}$ and the trivial Parseval identity $\int_0^1 |S_{\mathcal{S}_2}|^2 = \#\mathcal{S}_2(2X) \ll X \log\log X/\log X$ give

$$\sum_{X<N\le 2X} \big|R_2(N) - \mathrm{main}(N)\big|^2 \;\ll\; X^3 \log\log X\, (\log X)^{-2A-1}.$$

Proof. The first inequality is Bessel: $R_2(N) - \mathrm{main}(N) = \int_{\mathfrak{m}} S_{\mathbb{P}} S_{\mathcal{S}_2} e(-N\alpha)$ are Fourier coefficients of $S_{\mathbb{P}} S_{\mathcal{S}_2} \mathbf{1}_{\mathfrak{m}}$, so their ℓ^2 sum is at most $\int_{\mathfrak{m}} |S_{\mathbb{P}} S_{\mathcal{S}_2}|^2$. The second bounds one factor by its sup on $\mathfrak{m}$ and integrates the other; $\int_0^1 |S_{\mathcal{S}_2}|^2 = \#\mathcal{S}_2(2X)$ since cross terms cancel. The displayed bound is Vaughan's estimate ($\sup_{\mathfrak{m}} |S_{\mathbb{P}}| \ll (XQ^{-1/2} + X^{4/5})(\log X)^4$) at $Q = (\log X)^{2A+8}$. □

The only arithmetic input is the *classical* Vinogradov–Vaughan bound for the *prime* sum; the semiprime sum enters through its L^2 norm alone—no bilinear or sieve estimate for $S_{\mathcal{S}_2}$ is required.

Lemma 4 (Uniform lower bound for the blocked channel sum)**.** *Fix $0 < \eta < 1/2$. Uniformly for even $N \in (X, 2X]$, the truncated blocked channel sum*

$$W_f(N; X^\eta) := \sum_{\substack{3\le r\le X^\eta \\ r\in\mathbb{P},\ r\nmid N}} \frac{r-1}{r-2}\,\frac{1}{r}\,\frac{\log N}{\log(N/r)} \;\gg_\eta\; \log\log X,$$

and consequently $W_f(N) \gg_\eta \log\log X$.

Proof. For $r \leq X^\eta$ and $N \in (X, 2X]$ the archimedean channel weight satisfies $w_r(N) = I_r(N)/I_1(N) \gg_\eta 1/r$, so $W_f(N; X^\eta) \gg_\eta \sum_{3 \leq r \leq X^\eta,\, r \nmid N} 1/r$. By Mertens' theorem $\sum_{3 \leq r \leq X^\eta} 1/r = \log\log X + O_\eta(1)$. The reciprocal mass removed by the primes $r \mid N$ is $O(\log\log\log X)$: writing $P_z = \prod_{3 \leq p \leq z} p$, the bound $P_z \leq 2X$ forces $z \ll \log X$, so $\sum_{r \mid N} 1/r \leq \sum_{3 \leq r \ll \log X} 1/r = O(\log\log\log X)$. Hence $\sum_{3 \leq r \leq X^\eta,\, r \nmid N} 1/r \geq \log\log X - O(\log\log\log X) \gg \log\log X$. $\square$

Lemma 5 (Large-r channels are negligible almost everywhere)**.** *Fix $0 < \eta < 1/2$ and let $T_\eta(N)$ count the representations $N = p + rs$ with p, r, s prime, $r \leq s$, and $r > X^\eta$. Then for every $\varepsilon > 0$,*

$$\#\Big\{X < N \leq 2X,\ 2 \mid N :\ T_\eta(N) > \varepsilon\, \tfrac{X \log\log X}{\log^2 X}\Big\} = o(X).$$

Proof. A first-moment count gives $\sum_{X < N \leq 2X} T_\eta(N) \ll \frac{X^2}{\log^2 X} \sum_{X^\eta < r \leq \sqrt{2X}} \frac{1}{r}$, and $\sum_{X^\eta < r \leq \sqrt{X}} 1/r = \log\log\sqrt{X} - \log\log X^\eta + o(1) = \log\frac{1}{2\eta} + o(1)$ by Mertens. Hence the mean of T_η over the $\sim X/2$ even $N \in (X, 2X]$ is $\ll_\eta X/\log^2 X = o(X \log\log X/\log^2 X)$, and Markov's inequality gives the claim. $\square$

Proposition 3 (Major-arc evaluation for the truncated semiprime channel)**.** *Fix $0 < \eta < 1/2$, put $Y = X^\eta$, and define $R_2^{(\leq Y)}(N) := \sum_{p+rs=N,\ p,r,s \in \mathbb{P},\ 3 \leq r \leq Y} 1$, with major-arc contribution $M^{(\leq Y)}(N)$. Then the standard major-arc calculation—the distribution of semiprimes in progressions of moduli $q \leq Q$ (§A.12, Lemma 6) together with Lemma 1—gives*

$$M^{(\leq Y)}(N) = 2C_2\, \mathfrak{S}(N) \sum_{\substack{3 \leq r \leq Y \\ r \in \mathbb{P},\ r \nmid N}} \frac{r-1}{r-2}\, I_r(N) + o\Big(\frac{X \log\log X}{\log^2 X}\Big)$$

for even $N \in (X, 2X]$, uniformly outside an admissible $o(X)$ exceptional set.

Theorem 3 (Linear almost-all asymptotic for the semiprime channel)**.** *Let $X \to \infty$ and let N range over even integers in $(X, 2X]$. Split the circle $[0,1] = \mathfrak{M} \cup \mathfrak{m}$ at denominator bound $Q = (\log X)^{2A+8}$ with $A > 2$ fixed, and put* $\mathrm{main}(N) = \int_{\mathfrak{M}} S_{\mathbb{P}}(\alpha) S_{\mathcal{S}_2}(\alpha) e(-N\alpha)\, d\alpha$. *Assume the major-arc evaluation of Proposition 3 (semiprimes in progressions of moduli $q \leq Q$). Then, for all but $o(X)$ even $N \in (X, 2X]$,*

$$R_2(N) = \mathrm{main}(N)(1 + o(1)),$$

and

$$\mathrm{main}(N) = 2C_2 \mathfrak{S}(N) \sum_{\substack{3 \leq r \leq \sqrt{N} \\ r \in \mathbb{P},\ r \nmid N}} \frac{r-1}{r-2} I_r(N)(1 + o(1)) = W_f(N)\, [2C_2 \mathfrak{S}(N) I_1(N)](1 + o(1)).$$

Consequently, using the Montgomery–Vaughan almost-all asymptotic for R_1,

$$\frac{R_2(N)}{R_1(N)} = W_f(N)(1 + o(1)) \qquad \textit{for all but } o(X) \textit{ even } N \in (X, 2X].$$

Proof. On the minor arcs, $R_2(N) - \mathrm{main}(N) = \int_{\mathfrak{m}} S_{\mathbb{P}}(\alpha) S_{\mathcal{S}_2}(\alpha) e(-N\alpha)\, d\alpha$, so by Proposition 2

$$\sum_{X < N \leq 2X} |R_2(N) - \mathrm{main}(N)|^2 \leq \int_{\mathfrak{m}} |S_{\mathbb{P}}(\alpha)|^2 |S_{\mathcal{S}_2}(\alpha)|^2\, d\alpha \leq \Big(\sup_{\alpha \in \mathfrak{m}} |S_{\mathbb{P}}(\alpha)|\Big)^2 \int_0^1 |S_{\mathcal{S}_2}(\alpha)|^2\, d\alpha \ll_A X^3 (\log\log X)(\log$$

By Lemma 4 and the major-arc evaluation, $\mathrm{main}(N) \gg X \log\log X/\log^2 X$ uniformly off an $o(X)$ set. Hence by Chebyshev the exceptional set $\mathcal{E}_\varepsilon(X) = \{X < N \leq 2X : |R_2(N) - \mathrm{main}(N)| > \varepsilon\, \mathrm{main}(N)\}$ satisfies

$$\#\mathcal{E}_\varepsilon(X) \ll_{\varepsilon,A} \frac{X^3 (\log\log X)(\log X)^{-2A-1}}{X^2 (\log\log X)^2 (\log X)^{-4}} \ll_{\varepsilon,A} \frac{X (\log X)^{3-2A}}{\log\log X} = o(X),$$

so $R_2(N) = \mathrm{main}(N)(1+o(1))$ off $\mathcal{E}_\varepsilon$. The channels $r > X^\eta$ contribute $o(\mathrm{main})$ to the actual count off a further $o(X)$ set (Lemma 5); their contribution to the major-arc main term is also $o(\mathrm{main})$, deterministically, since $\sum_{X^\eta < r \le \sqrt{N}} I_r(N)/I_1(N) \ll_\eta 1 = o(W_f(N; X^\eta))$. Hence the major-arc main term reduces to the truncated one of Proposition 3; with Lemma 1 this is $2C_2\mathfrak{S}(N)\sum_{r\nmid N} \frac{r-1}{r-2} I_r(N)(1+o(1)) = W_f(N)[2C_2\mathfrak{S}(N)I_1(N)](1+o(1))$. Intersecting with the full-density set on which $R_1(N) = 2C_2\mathfrak{S}(N)I_1(N)(1+o(1))$ (Montgomery–Vaughan [7]) yields $R_2/R_1 = W_f(N)(1+o(1))$ off $o(X)$. □

Lower-tail hypothesis. For every $\delta > 0$,

$$\frac{1}{X}\#\Big\{X < N \le 2X,\ 2 \mid N:\ \frac{R_2(N)}{R_1(N)} \le \delta\, W_f(N)\Big\} \longrightarrow 0 \qquad (\delta \downarrow 0,\ X \to \infty).$$

This is a one-sided control of the lower tail of R_2/R_1; it is strictly weaker than the full asymptotic but is *not* implied by the L^2 bound of Proposition 2.

Corollary 1 ($\beta_2 \to 1$, conditional on the lower-tail hypothesis)**.** *Under the lower-tail hypothesis above, the exceptional set $\mathcal{E}_X$ of Theorem 3 contributes $o(1)$ to the covariance* (26)*; the raw slope m_2 is then computed off $\mathcal{E}_X$, where $R_2/R_1 = W_f(1+o(1))$, so $m_2 \to 0$ and $\beta_2(X) \to 1$. Without this hypothesis, the second-moment result of Theorem 3 controls $R_2 - \mathrm{main}$, not the untruncated log-regression; for unconditional statements one restricts to truncated logarithms (§A.7).*

Remark 6 (The honest status of the limit)**.** *Theorem 3 is the linear statement obtained within the stated bounded reduction (the Vaughan–Parseval second moment and the major-arc input); Corollary 1 is what is usually quoted as "$\beta_2 \to 1$ for almost all N," but it carries a genuine extra hypothesis. The gap is precisely the lower tail of R_2: L^2 concentration of $R_2 - \mathrm{main}$ bounds how often R_2 is* large, *but the log-regression that defines β_2 also weights N where R_2 is* small, *and a sparse set with R_2 very small could in principle skew the covariance. The required one-sided lower-tail bound does not come from Chen's theorem, and we record* why *the natural attempt fails. Chen's method lower-bounds the* combined *count $R_{\le 2} = R_1 + R_2$, not the semiprime-only R_2; one would therefore try to extract $R_2 = R_{\le 2} - R_1$ from a Chen-type lower bound on $R_{\le 2}$ and the Montgomery–Vaughan value of R_1 [7]. Carrying this out, it does* not *close: the unconditional Chen-type lower bound has the shape $R_{\le 2}(N) \ge c\, 2C_2\mathfrak{S}(N)N/\log^2 N$ with $c < 1$, i.e. at the* scale of R_1*, whereas the semiprime main term is* $\mathrm{main}(N) \asymp 2C_2\mathfrak{S}(N)\, W_f(N)\, N/\log^2 N$ *with $W_f \sim \log\log N$. Subtracting $R_1 = (1+o(1))\, 2C_2\mathfrak{S}(N)N/\log^2 N$ from a quantity a factor* $\log\log N$ smaller *is vacuous; it does not even yield $R_2 \gg R_1$. The genuine obstruction is thus a lower bound on the* semiprime channel *at the correct* $\log\log N$ *scale (capturing W_f), which the sieve does not currently supply, and we leave it open. The hypothesis is amply confirmed empirically—the exceptional set is* empty *at the* 5% *level up to $5 \cdot 10^8$ (Table 6), so in the computed range no N has R_2/R_1 even near the tail—but we flag it rather than claim it.*

Remark 7 (What is and is not proved)**.** *Theorem 3 is stated as a* bounded reduction, *not as a self-contained unconditional theorem. The minor-arc variance estimate (Proposition 2) is unconditional; the identification of the major-arc main term is the standard Hardy–Littlewood major-arc evaluation for the channel forms s and $N - rs$ (Proposition 3, §A.12), which we invoke but do not write out in full—it follows the prime distribution by Selberg–Sathé convolution [10, 11]. Every ingredient—Vaughan's minor-arc bound [13], Siegel–Walfisz, Montgomery–Vaughan, and the Euler-product identity Lemma 1—is classical, and the one thing we make* explicit *is the singular-series cancellation of Lemma 1. This is in sharp contrast to the pointwise Assumption 1, which lies beyond current technology, and to the log-regression statement $\beta_2 \to 1$, which needs the lower-tail hypothesis of Corollary 1.*

Remark 8 (Relation to the literature)**.** *We stress that Theorem 3 is not a new estimate (a novelty claim about the* reduction, *to be kept distinct from the novelty claim about the* identity *in*

Remark 3). The exceptional-set bound it rests on is the Montgomery–Vaughan template for the binary Goldbach asymptotic [7], and the only departure—that the semiprime sum enters through its L^2 norm alone, with no bilinear or sieve input—makes the argument, if anything, easier *than for R_1. Strong unconditional results on Goldbach-type counts, and the $(1+a)$ interpolation between Goldbach and Chen [6], already operate in this regime; we claim no improvement on them, and the linear almost-all asymptotic for R_2 should be regarded as a standard consequence of the classical circle-method framework, under the stated major-arc input. What is specific here is the singular-series cancellation of Lemma 1, which exhibits* $\mathrm{main}(N) = 2C_2\,\mathfrak{S}(N)\,W_f(N)\,J(N)$ *explicitly and is the reason $\mathfrak{S}$ disappears from the ratio R_2/R_1. The contribution of this appendix is the explicit bookkeeping and its consequences for the effective exponent β_2, not the almost-all asymptotic itself.*

A.7 Truncated logarithms and the blocked-channel model

The second-moment estimate of §A.6 controls $R_2 - \mathrm{main}$ in L^2, not $\log(R_2/R_1)$, and Corollary 1 flagged the lower tail as the obstruction to the untruncated slope. We add two short results that delimit the gap exactly: the second moment *does* control a *truncated* logarithmic statistic (Proposition 4), and an idealized blocked-channel model proves the $1/\log\log X$ deficit (Proposition 5). Neither establishes the untruncated $\beta_2 \to 1$; together they show that only the lower tail, on a sparse set, is left open.

A truncated-log bridge. Write $\rho(N) = R_2(N)/\mathrm{main}(N)$ and, on a dyadic window $\mathcal{W} = \{N \in (X, 2X] : 2 \mid N\}$, let

$$\eta_X := \frac{1}{|\mathcal{W}|}\sum_{N\in\mathcal{W}} \left|\rho(N) - 1\right|^2,$$

the relative variance bounded in Proposition 2/Theorem 3: $\eta_X \ll (\log X)^{3-2A}/\log\log X \to 0$ for $A \geq 2$. Fix a truncation level $\varepsilon_X = (\log X)^{-A'}$ ($A' > 0$ fixed) and set $L_X(N) = \log\max\{\rho(N), \varepsilon_X\}$, the response with its lower tail clipped.

Proposition 4 (Truncated logarithmic stability)**.** *If $\eta_X \log(1/\varepsilon_X) \to 0$ (which holds for the above ε_X whenever $A \geq 2$), then*

$$\frac{1}{|\mathcal{W}|}\sum_{N\in\mathcal{W}} |L_X(N)| \;\ll\; \left(\eta_X \log(1/\varepsilon_X)\right)^{1/3} = o(1), \qquad \frac{1}{|\mathcal{W}|}\sum_{N\in\mathcal{W}} L_X(N)^2 = o(1).$$

Proof. Let $\delta_X \in (0, \frac{1}{2})$. On the good set $G = \{|\rho - 1| \leq \delta_X\}$ we have $\rho \geq 1 - \delta_X > \varepsilon_X$ for large X, so $L_X = \log\rho$ and $|L_X| \leq 2\delta_X$. By Chebyshev the complement has density $\leq \eta_X/\delta_X^2$; there, if $\rho < 1$ then $\max\{\rho, \varepsilon_X\} \geq \varepsilon_X$ gives $|L_X| \leq \log(1/\varepsilon_X)$, while if $\rho \geq 1$ then $R_2(N) \leq N$ and $\mathrm{main}(N) \gg X\log\log X/\log^2 X$ on $\mathcal{W}$ give $\rho \ll \log^2 X/\log\log X$, hence $|L_X| = \log\rho \leq 2\log\log X + O(1) \leq C\log(1/\varepsilon_X)$. Therefore $|\mathcal{W}|^{-1}\sum_{\mathcal{W}} |L_X| \leq 2\delta_X + C\,\eta_X\delta_X^{-2}\log(1/\varepsilon_X)$, and $\delta_X = (\eta_X\log(1/\varepsilon_X))^{1/3}$ yields the first bound. The second follows from the pointwise $|L_X| \leq C\log(1/\varepsilon_X)$: $|\mathcal{W}|^{-1}\sum L_X^2 \leq C\log(1/\varepsilon_X)\,|\mathcal{W}|^{-1}\sum|L_X| \to 0$ for $A \geq 2$. $\square$

Since $\log(R_2/R_1) = \log\rho + \log(\mathrm{main}/R_1)$ and $\mathrm{main}(N)/R_1(N) = W_f(N)(1 + o(1))$ off a set of size $O(X^{1-\delta})$ (the major arcs of §A.12 and Montgomery–Vaughan for R_1), Proposition 4 shows that the $\log\rho$ part contributes $o(1)$ to both the variance and the covariance of the β_2-regression once truncated. The *truncated* log-regression slope is therefore governed entirely by the channel term $\log W_f$, treated next. This is the precise sense in which the second-moment machinery already suffices: the only quantity it cannot reach is the untruncated lower tail, which is empirically inactive ($\mathrm{CV}(b) < 0.7\%$ up to $5 \cdot 10^8$, Table 6), so in the computed range the truncated and untruncated slopes coincide.

The blocked-channel model. We make the mechanism behind $\beta_2(X) = 1 - c/W_{f,0}$ (Theorem 1) precise in the Kubilius-type model of Lemma 2, where the indicators $e_\ell = \mathbf{1}[\ell \mid N]$ are independent Bernoulli$(1/\ell)$ over distinct odd primes. Freezing the weights at scale X, with $a_r = \frac{r-1}{r-2}\frac{1}{r}$ set

$$W_{0,X} = \sum_{3 \le r \le \sqrt{X}} a_r, \qquad D_X(N) = \sum_{\substack{r \mid N \\ 3 \le r \le \sqrt{X}}} a_r, \qquad W_X(N) = W_{0,X} - D_X(N),$$

and, with $g_\ell = \log \frac{\ell-1}{\ell-2}$, $Y_X(N) = \sum_{\ell \mid N,\, \ell > 2} g_\ell$ (the model's $\log \mathfrak{S}$).

Proposition 5 (Blocked-channel slope; model)**.** *In the independent-divisibility model,*

$$\mathrm{Cov}(\log W_X,\, Y_X) = -\frac{A_X}{W_{0,X}} + O(W_{0,X}^{-2}), \quad A_X = \sum_{3 \le r \le \sqrt{X}} a_r\, g_r\, \tfrac{1}{r}(1 - \tfrac{1}{r}) > 0,$$

and $\mathrm{Var}(Y_X) = B_X := \sum_{2 < \ell \le \sqrt{X}} g_\ell^2\, \frac{1}{\ell}(1 - \frac{1}{\ell}) \to B$*. Hence the model slope satisfies*

$$\beta_{2,X}^{\mathrm{model}} = 1 + \frac{\mathrm{Cov}(\log W_X, Y_X)}{\mathrm{Var}(Y_X)} = 1 - \frac{C_X}{W_{0,X}} + O(W_{0,X}^{-2}), \qquad C_X = \frac{A_X}{B_X},$$

with A_X, B_X positive and bounded; since $W_{0,X} = \log\log X + O(1)$ (Mertens),

$$\boxed{1 - \beta_{2,X}^{\mathrm{model}} \asymp \frac{1}{\log\log X}.}$$

Proof. As $\mathbb{E}[D_X] = \sum_r a_r/r = O(1)$ while $W_{0,X} \to \infty$, the event $\{D_X \le \frac{1}{2}W_{0,X}\}$ has complementary probability $O(W_{0,X}^{-2})$ (Chebyshev), and there $\log W_X = \log W_{0,X} - D_X/W_{0,X} + O(D_X^2/W_{0,X}^2)$; hence $\mathrm{Cov}(\log W_X, Y_X) = -W_{0,X}^{-1}\,\mathrm{Cov}(D_X, Y_X) + O(W_{0,X}^{-2})$. By independence $\mathrm{Cov}(e_r, e_\ell) = \delta_{r\ell}\,\frac{1}{r}(1 - \frac{1}{r})$, so $\mathrm{Cov}(D_X, Y_X) = \sum_r a_r g_r \frac{1}{r}(1 - \frac{1}{r}) = A_X$ and $\mathrm{Var}(Y_X) = B_X$; both series converge ($a_r g_r/r \asymp r^{-3}$, $g_\ell^2/\ell \asymp \ell^{-3}$), and $W_{0,X} = \sum_{r \le \sqrt{X}} \frac{r-1}{(r-2)r} = \sum_{r \le \sqrt{X}} \frac{1}{r} + O(1) = \log\log X + O(1)$. □

This model theorem isolates the central mechanism: each unblocked semiprime channel inherits $\mathfrak{S}(N)$ in full, but the same small-prime divisibility that *raises* $\mathfrak{S}(N)$ (through $g_\ell > 0$) *removes* the matching r-channel from the Mertens sum (through $-a_r$ in D_X). The covariance is therefore negative, purely mechanical, and multi-channel—a diagonal sum over $r = \ell$ with no dominant term, consistent with the $\sim 10\%$ weight of the $3 \mid N$ atom in §7.2—and of size $1/W_{0,X} \sim 1/\log\log X$. It makes the constant $c = A/B$ of Theorem 1 precise *within the model*; it is not a theorem about the primes, for which the same identity needs the unproven uniformity of Assumption 1.

Remark 9 (Chen-rescue threshold; heuristic)**.** *Model the number of balanced Goldbach pairs in the band $p \in [N/2 - k,\, N/2]$ as Poisson with mean $\lambda_k(N) \approx 2C_2\,\mathfrak{S}(N)\,k/\log^2(N/2)$ (the Hardy–Littlewood density on a band of k integers). The chance of no prime pair is then $e^{-\lambda_k}$, so the fraction of N that require a semiprime rises as the band narrows and crosses one half near $\lambda_k \approx \log 2$, i.e. at*

$$k_\star(N) \approx \frac{\log^2 N}{2C_2\,\mathfrak{S}(N)}.$$

This reproduces both features observed in §7.5: the peak band-scale grows like $\log^2 N$, and a large $\mathfrak{S}(N)$ pushes the threshold to smaller *k (primes are locally abundant, so balance must be tighter before semiprimes are needed). The Poisson step is heuristic—it ignores correlations among nearby pairs.*

A.8 The Ω-generating function and singular-series response

The layer experiment of §7.3 can be organized by a single generating function. For an even integer N, define

$$Z_N(z) := \sum_{p<N} \mathbf{1}_{\mathbb{P}}(p)\, z^{\Omega(N-p)}.$$

Then $[z^k]Z_N(z)$ is the number of representations $N = p + q$ with p prime and $\Omega(q) = k$; in particular $[z]Z_N(z) = R_1(N)$ and $[z^2]Z_N(z) = R_2(N)$. For $z = e^{-\beta}$ this is the one-sided arithmetic partition function for the Ω-energy of the second summand (its two-sided analogue is the energy reframing of Appendix B).

The following is a *model* statement, not a theorem about the primes: it is exact in the local blocked-channel model of §A.7, and it explains why a large singular series pushes the second summand toward lower arithmetic complexity. Fix y and let $\mathcal{P}_y$ be the odd primes $\ell \le y$; model $B_\ell := \mathbf{1}_{\ell|N}$ as independent Bernoulli$(1/\ell)$ (Lemma 2). With $g_\ell = \log \frac{\ell-1}{\ell-2}$ as before, the model $\log \mathfrak{S}$ is $Y_y(N) := \sum_{\ell\in\mathcal{P}_y} g_\ell B_\ell$. Given the divisibility vector $B = (B_\ell)$, the admissible second summand $q = N - p$ has local divisibility probabilities

$$\Pr(\ell \mid q \mid B_\ell) = \pi_\ell(B_\ell) := \begin{cases} 0, & B_\ell = 1, \\ \dfrac{1}{\ell-1}, & B_\ell = 0, \end{cases}$$

since $\ell \mid N$ blocks the channel ($q \equiv -p \not\equiv 0$ for admissible $p \not\equiv 0 \pmod{\ell}$), whereas if $\ell \nmid N$ exactly one admissible residue of p gives $q \equiv 0 \pmod{\ell}$. Let

$$F_y(z; B) := \prod_{\ell\in\mathcal{P}_y} (1 + (z-1)\pi_\ell(B_\ell)) = \prod_{\ell\in\mathcal{P}_y} \left(1 + (z-1)\frac{1-B_\ell}{\ell-1}\right)$$

be the local Ω-generating function of q after the inherited Goldbach singular factor is divided out.

Proposition 6 (Generating-function response; model)**.** *Define the model singular response $\beta_y(z) - 1 := \mathrm{Cov}(\log F_y(z; B), Y_y(B))/\mathrm{Var}(Y_y(B))$. Then, for every fixed $z > 0$ for which the logarithms are defined,*

$$\beta_y(z) - 1 = -\frac{\sum_{\ell\in\mathcal{P}_y} \frac{1}{\ell}(1-\frac{1}{\ell})\, g_\ell \log(1 + \frac{z-1}{\ell-1})}{\sum_{\ell\in\mathcal{P}_y} \frac{1}{\ell}(1-\frac{1}{\ell})\, g_\ell^2}.$$

Consequently $\beta_y(z) < 1$ for $z > 1$, $\beta_y(z) > 1$ for $0 < z < 1$, and $\beta_y(1) = 1$.

Proof. Write $\log F_y(z; B) = \sum_\ell (1 - B_\ell)\, h_\ell(z)$ with $h_\ell(z) := \log(1 + \frac{z-1}{\ell-1})$, and $Y_y(B) = \sum_\ell g_\ell B_\ell$. By independence the off-diagonal covariances vanish, so

$$\mathrm{Cov}(\log F_y, Y_y) = \sum_\ell h_\ell(z)\, g_\ell\, \mathrm{Cov}(1 - B_\ell, B_\ell) = -\sum_\ell h_\ell(z)\, g_\ell\, \tfrac{1}{\ell}(1 - \tfrac{1}{\ell}),$$

and $\mathrm{Var}(Y_y) = \sum_\ell g_\ell^2\, \frac{1}{\ell}(1-\frac{1}{\ell})$. Dividing gives the formula. As $g_\ell > 0$ for every odd prime, the sign is that of $h_\ell(z)$: positive for $z > 1$, negative for $0 < z < 1$, zero at $z = 1$. □

The singular series therefore acts on the whole Ω-generating function, not only on the isolated layers R_1, R_2: weights $z > 1$ overweight high complexity, where the response falls below one (large $\mathfrak{S}$ suppresses the high-Ω tail), and $0 < z < 1$ overweight low complexity, where it rises above one (large $\mathfrak{S}$ favours low-Ω summands). The sign reversal of §7.3 is the coefficient-level shadow of this single response.

Stochastic-dominance form. Let $\Omega_y(q) := \sum_{\ell\in\mathcal{P}_y} \mathbf{1}_{\ell|q}$ in the same model; conditional on B its summands are independent Bernoulli$(\pi_\ell(B_\ell))$. If $B' \geq B$ coordinatewise then $\pi_\ell(B'_\ell) \leq \pi_\ell(B_\ell)$ for every ℓ, so $\Omega_y(q) \mid B' \preceq_{\mathrm{st}} \Omega_y(q) \mid B$: adding blockers shifts the factor-count law to the left. Since $Y_y(B)$ is increasing in B, for any increasing φ the map $B \mapsto \mathbb{E}[\varphi(\Omega_y(q)) \mid B]$ is decreasing, and the Harris–FKG inequality for product measures gives $\operatorname{Cov}\left(\mathbb{E}[\varphi(\Omega_y(q)) \mid B],\, Y_y(B)\right) \leq 0$. Taking $\varphi = \mathbf{1}_{\{m\geq k\}}$ shows the high-complexity tail is negatively correlated with the singular series; $\varphi = \mathbf{1}_{\{m\leq k\}}$ gives the positive correlation for low-complexity events.

Poissonized layers. With $\mu_y(B) := \sum_{\ell\in\mathcal{P}_y} \pi_\ell(B_\ell) = \sum_\ell \frac{1-B_\ell}{\ell-1}$,

$$\operatorname{Cov}(\mu_y, Y_y) = -\sum_{\ell\in\mathcal{P}_y} \frac{g_\ell}{\ell-1}\,\tfrac{1}{\ell}\big(1-\tfrac{1}{\ell}\big) < 0.$$

Approximating $\Omega_y(q) \mid B$ by a Poisson variable of mean $\mu_y(B)$ and linearizing

$$\log \Pr(\Omega_y(q) = k \mid B) = \text{const} + \Big(\frac{k}{\bar{\mu}_y} - 1\Big)\big(\mu_y(B) - \bar{\mu}_y\big) + \cdots$$

about $\bar{\mu}_y = \mathbb{E}\mu_y$ shows, since $\operatorname{Cov}(\mu_y, Y_y) < 0$, a positive singular-series response for layers below the typical factor count ($k < \bar{\mu}_y$) and a negative one above it ($k > \bar{\mu}_y$). This is the mechanism behind the transition from positive response in the Goldbach and Chen layers to negative response for higher Ω-layers; it gives the direction and the sign reversal, not the exact numerical β_k.

A.9 Weak uniformity of the location measures

The experiments of §7.6 suggest that the representation clouds become continuous after normalization: the empirical laws of q/N for both prime and semiprime q approach the uniform law on $(0,1)$, up to small endpoint effects. This is what the weighted Hardy–Littlewood heuristic predicts. For a bounded C^1 test function $\varphi : [0,1] \to \mathbb{R}$ write, when $R_t(N) \neq 0$,

$$\mu_N^{(1)}(\varphi) := \frac{1}{R_1(N)} \sum_{\substack{p<N\\ p,\,N-p\in\mathbb{P}}} \varphi\Big(\tfrac{N-p}{N}\Big), \qquad \mu_N^{(2)}(\varphi) := \frac{1}{R_2(N)} \sum_{\substack{p<N\\ p\in\mathbb{P},\,N-p\in\mathcal{S}_2}} \varphi\Big(\tfrac{N-p}{N}\Big),$$

the averages of φ against the type-t location measures of §7.6.

Hypothesis 1 (Weighted prime and semiprime channel asymptotics)**.** *For every bounded C^1 test function φ and almost all even N,*

$$\sum_{\substack{p<N\\ p,\,N-p\in\mathbb{P}}} \varphi\Big(\tfrac{N-p}{N}\Big) = (1+o(1))\, 2C_2\, \mathfrak{S}(N)\, \frac{N}{\log^2 N} \int_0^1 \varphi(x)\, dx,$$

and, for each unblocked semiprime channel $q = rs$, uniformly in the range of r used in the paper,

$$\sum_{\substack{s\in\mathbb{P}\\ N-rs\in\mathbb{P}}} \varphi\Big(\tfrac{rs}{N}\Big) = (1+o(1))\, 2C_2\, \mathfrak{S}(N)\, \frac{r-1}{r-2}\, \frac{N}{r\log^2 N} \int_0^1 \varphi(x)\, dx,$$

the channel omitted when $r \mid N$.

Proposition 7 (Weak uniformity of the location measures; conditional)**.** *Under Hypothesis 1, for almost all even N,*

$$\mu_N^{(1)}(\varphi) = \int_0^1 \varphi(x)\, dx + o(1), \qquad \mu_N^{(2)}(\varphi) = \int_0^1 \varphi(x)\, dx + o(1).$$

Equivalently, the empirical measures of q/N in the prime and semiprime channels converge weakly to the uniform law on $(0,1)$, in the same smoothed/almost-all sense as the weighted asymptotics.

Proof. For the prime channel, the first weighted asymptotic with $\varphi \equiv 1$ gives $R_1(N) = (1 + o(1))\, 2C_2\mathfrak{S}(N)N/\log^2 N$; dividing the two estimates yields $\mu_N^{(1)}(\varphi) = \int_0^1 \varphi + o(1)$. For the semiprime channel, writing $q = rs$ and summing the per-channel asymptotic over the unblocked $r \le \sqrt{N}$,

$$\sum_{\substack{p<N \\ p\in\mathbb{P},\, N-p\in\mathcal{S}_2}} \varphi\Big(\tfrac{N-p}{N}\Big) = (1+o(1))\, 2C_2\mathfrak{S}(N)\, \frac{N}{\log^2 N}\Big(\sum_{\substack{3\le r\le\sqrt{N} \\ r\nmid N}} \frac{r-1}{r-2}\frac{1}{r}\Big)\int_0^1 \varphi(x)\, dx,$$

the parenthesis being the Mertens channel sum $W_f(N)$. Taking $\varphi \equiv 1$ gives the matching asymptotic for $R_2(N)$; dividing cancels both the inherited singular series $\mathfrak{S}(N)$ and the channel sum, leaving $\mu_N^{(2)}(\varphi) = \int_0^1 \varphi + o(1)$. This is weak convergence against every bounded C^1 test function. □

This explains why the discontinuous pointwise representation sets nonetheless produce smooth limiting clouds: the singular series governs the local *density* of representations, but after normalizing each measure it cancels—together with the Mertens channel sum in the semiprime case—leaving the uniform spatial law. It is the conditional counterpart of the weak-continuity diagnostics of §7.6, and like the other appendix results it is conditional on the stated weighted asymptotics, not an unconditional theorem about primes.

A.10 The least-unblocked-prime law and residual collapse

The residual exponent of §7.7 suggests that the Chen surplus is not merely large: for most integers it collapses to the first locally admissible semiprime channel. This can be formalized under the same kind of fixed-channel almost-all input that underlies the circle-method discussion above.

For an even integer N, define the nondegenerate minimal multiplier

$$r_\star(N) := \min\{r \ge 3 :\ r \in \mathbb{P},\ \exists p, s \in \mathbb{P},\ N = p + rs\}.$$

(In §7.7 the multiplier ranges over all primes; we restrict to $r \ge 3$ to exclude the parity-degenerate channel $r = 2$, which forces $p = N - 2s$ even, hence $p = 2$ and $s = N/2 - 1$ prime—a density-zero family. When $r \mid N$ the congruence $p = N - rs \equiv 0 \pmod r$ likewise forces $p = r$, a density-zero exception for each fixed r; so the two definitions agree for almost all N.) Let

$$\rho(N) := \min\{r \ge 3 :\ r \in \mathbb{P},\ r \nmid N\}$$

be the least locally unblocked odd prime. We use the following fixed-channel almost-all hypothesis—an averaged, single-channel form of Assumption 1.

Hypothesis 2 (Fixed-channel almost-all asymptotic)**.** *For every fixed finite set $\mathcal{R}$ of odd primes and every fixed $0 < \eta < 1/2$, for all but $o_{\mathcal{R},\eta}(X)$ even $N \le X$, uniformly for every $r \in \mathcal{R}$ with $r \nmid N$,*

$$C_{r,\eta}(N) := \#\Big\{s \in \mathbb{P} :\ \eta\tfrac{N}{r} \le s \le (1-\eta)\tfrac{N}{r},\ \ N-rs \in \mathbb{P}\Big\} = (1+o(1))\, 2C_2\, \mathfrak{S}(N)\, \frac{r-1}{r-2}\, \frac{(1-2\eta)N}{r\log^2 N}.$$

In particular, for every fixed unblocked r the channel $N = p + rs$ contains a representation with $s \asymp N/r$ for almost all even N.

Theorem 4 (Least-unblocked-prime law; conditional)**.** *Assume Hypothesis 2. Then* $r_\star(N) = \rho(N)$ *for almost all even integers* N*, and consequently* $a_{\mathrm{Res}}(N) \to 1$ *for almost all even integers* N*.*

Proof. Fix a finite set of odd primes $\mathcal{R}_Y := \{3 \le r \le Y :\ r \in \mathbb{P}\}$, and let $\rho_Y(N)$ be the least prime in $\mathcal{R}_Y$ not dividing N, when one exists. Consider an even $N \le X$ for which $\rho_Y(N)$ exists. Every odd prime $r < \rho_Y(N)$ in $\mathcal{R}_Y$ divides N, so its channel is locally blocked: $p = N - rs \equiv 0 \pmod{r}$, whence p is prime only if $p = r$, forcing $s = N/r - 1$. For fixed r the set of $N \le X$ with $N/r - 1$ prime has size $O(X/(r \log X)) = o(X)$; the finite union over $r \in \mathcal{R}_Y$ is exceptional of size $o_Y(X)$, and outside it no blocked $r < \rho_Y(N)$ yields a representation.

On the other hand $\rho_Y(N) \nmid N$, so by Hypothesis 2 applied to $\mathcal{R}_Y$, for all but $o_Y(X)$ even $N \le X$ the channel $r = \rho_Y(N)$ carries a representation $N = p + \rho_Y(N)s$ with $s \asymp N/\rho_Y(N)$. Hence, outside a set of size $o_Y(X)$, $r_\star(N) = \rho_Y(N)$ whenever $\rho_Y(N)$ exists.

It remains to remove the cutoff Y. The even N for which $\rho_Y(N)$ does not exist are exactly those divisible by every odd prime $r \le Y$, of density $\prod_{3 \le r \le Y} \frac{1}{r}$ among even integers. Therefore

$$\limsup_{X\to\infty} \frac{\#\{N \le X \text{ even}:\ r_\star(N) \ne \rho(N)\}}{X/2} \le \prod_{\substack{3 \le r \le Y\\ r \in \mathbb{P}}} \frac{1}{r},$$

and the product tends to 0 as $Y \to \infty$, giving $r_\star(N) = \rho(N)$ for almost all even N. Finally, on the same density-one set the representation has $r = r_\star(N) \le Y$ (outside a set of arbitrarily small upper density) and $s \asymp N/r$, so

$$a_{\mathrm{Res}}(N) = 1 + \frac{\log r_\star(N)}{\log s} \le 1 + \frac{\log Y}{\log N + O_Y(1)},$$

which tends to 1 as $N \to \infty$ for each fixed Y; since the exceptional density can be made arbitrarily small, $a_{\mathrm{Res}}(N) \to 1$ for almost all even N. □

This gives a precise meaning to the empirical "residual collapse" of §7.7: the small multiplier is not mysterious in the model—it is usually just the first prime channel not locally blocked by divisibility of N. In the independent-divisibility model of Lemma 2 it predicts

$$\Pr(\rho(N) = 3) = \tfrac{2}{3}, \quad \Pr(\rho(N) = 5) = \tfrac{1}{3}\cdot\tfrac{4}{5} = \tfrac{4}{15}, \quad \Pr(\rho(N) = 7) = \tfrac{1}{3}\cdot\tfrac{1}{5}\cdot\tfrac{6}{7} = \tfrac{2}{35},$$

so $\Pr(\rho(N) \in \{3,5,7\}) = \frac{2}{3} + \frac{4}{15} + \frac{2}{35} \approx 0.9905$, explaining why a very small multiplier dictionary covers almost all integers in the experiments. It also clarifies the link to the singular series: the same small-prime divisibility that *increases* $\mathfrak{S}(N)$ *blocks* the corresponding small r-channels, so the least available Chen multiplier is governed by local obstructions. The statement is conditional on Hypothesis 2—a standard fixed-channel almost-all input—and is not an unconditional theorem about primes; granting that input, the residual collapse and the distribution of $r_\star$ follow from a clean local-obstruction argument.

A.11 A last-digit model for the carry excess

The carry geometry of §8.2 can be partly explained by a local digit calculation: prime-like summands are not uniform among terminal digits but restricted to reduced residue classes, and this restriction alone creates a positive carry bias in the least significant digit. Fix a base $b \ge 2$ and let $U_b := \{1 \le u \le b-1 : (u,b) = 1\}$ be the reduced residue digits modulo b. For digits $x, y \in \{0, \dots, b-1\}$ let $\kappa_b(x,y) := \mathbf{1}_{\{x+y \ge b\}}$ be the carry out of the least significant digit.

Proposition 8 (Local carry excess; model)**.** *Let* X, Y *be independent uniform digits in* $\{0, \dots, b-1\}$ *and* U, V *independent uniform digits in* U_b*. Then*

$$\mathbb{E}\,\kappa_b(U,V) = \tfrac{1}{2} + \tfrac{1}{2}\Pr(U + V = b), \qquad \mathbb{E}\,\kappa_b(X,Y) = \frac{b-1}{2b},$$

so $\mathbb{E}\,\kappa_b(U,V) - \mathbb{E}\,\kappa_b(X,Y) = \frac{1}{2b} + \frac{1}{2}\Pr(U+V=b) > 0$*: reduced-residue summands have a strictly larger least-digit carry probability than unrestricted ones.*

Proof. For unrestricted digits, $\#\{(x,y) : 0 \le x,y \le b-1,\ x+y \ge b\} = \sum_{x=1}^{b-1} x = \frac{b(b-1)}{2}$, so $\mathbb{E}\,\kappa_b(X,Y) = (b-1)/(2b)$. For reduced residues, the involution $(u,v) \mapsto (b-u, b-v)$ maps U_b^2 to itself (since $(u,b)=1 \iff (b-u,b)=1$), sends $\{u+v<b\}$ bijectively to $\{u+v>b\}$, and fixes $\{u+v=b\}$; hence those two strict sets are equinumerous and $\Pr(U+V \ge b) = \frac{1}{2} + \frac{1}{2}\Pr(U+V=b)$. □

The calculation is deliberately local: it does not prove the full carry statistics of actual Goldbach pairs, nor assume independence of higher digits. It shows that the most basic prime residue restriction already pushes the carry count up. In base 2 both odd prime summands end in 1, so the least digit always carries; in larger bases the effect survives after averaging over reduced classes. Since the total carry count obeys the Kummer–Legendre identity $\#\mathrm{carries}_b(a,q) = (S_b(a)+S_b(q)-S_b(N))/(b-1)$ (§8.2), the proposition is a positive *local* contribution to it, and the empirical excess of §8.2 reads as its multi-digit continuation.

Remark 10 (Fixed-N caveat)**.** *For a fixed terminal digit N* mod *b the carry bias can depend on the residue class and need not be positive in every class; the proposition is an averaged local-prime model, explaining why a positive excess is natural* on average *while the full statistic remains a global multi-digit measurement. Like the other appendix results it is model-based, not a theorem about the empirical carry law.*

A.12 The major arcs: semiprimes in progressions

Here we supply the one routine ingredient left open in Theorem 3: the evaluation of $\mathrm{main}(N) = \int_{\mathfrak{M}} S_{\mathbb{P}} S_{\mathcal{S}_2} e(-N\alpha)$, which rests on the distribution of semiprimes in progressions. Write $\pi_2(x;q,a) = \#\{n \le x : n \in \mathcal{S}_2,\ n \equiv a\ (q)\}$ and $\mathcal{S}_2^*(x;q) = \#\{n \le x : n \in \mathcal{S}_2,\ (n,q)=1\}$.

Lemma 6 (Selberg–Sathé in progressions)**.** *Fix $A \ge 1$. There is $c = c(A) > 0$ such that, uniformly for $q \le (\log x)^A$ and $(a,q)=1$,*

$$\pi_2(x;q,a) = \frac{\mathcal{S}_2^*(x;q)}{\phi(q)} + O_A\big(x\,e^{-c\sqrt{\log x}}\big).$$

Proof. Write each $n \in \mathcal{S}_2$ as $n = p_1p_2$ with $p_1 \le p_2$. For $(a,q)=1$ a contribution requires $(p_1p_2,q)=1$, so

$$\pi_2(x;q,a) = \sum_{\substack{p_1 \le \sqrt{x} \\ (p_1,q)=1}} \Big(\pi\big(\tfrac{x}{p_1}; q, a\bar{p}_1\big) - \pi\big(p_1; q, a\bar{p}_1\big)\Big), \qquad \bar{p}_1 = p_1^{-1} \pmod{q}.$$

Since $p_1 \le \sqrt{x}$ gives $\log(x/p_1) \ge \frac{1}{2}\log x$, the Siegel–Walfisz theorem applies uniformly for $q \le (\log x)^A$: $\pi(y;q,c) = \mathrm{li}(y)/\phi(q) + O(y\,e^{-c_1\sqrt{\log y}})$ for $(c,q)=1$. Summing the main terms gives $\frac{1}{\phi(q)}\sum_{p_1 \le \sqrt{x}, (p_1,q)=1}[\mathrm{li}(x/p_1) - \mathrm{li}(p_1)] = \frac{1}{\phi(q)}\mathcal{S}_2^*(x;q)$, the last identity being the same decomposition of $\mathcal{S}_2^*$ without the progression. The error is $\ll \sum_{p_1 \le \sqrt{x}} \frac{x}{p_1} e^{-c_1\sqrt{\log(x/p_1)}} \ll x\,e^{-c_2\sqrt{\log x}} \sum_{p_1 \le \sqrt{x}} \frac{1}{p_1} \ll x \log\log x\, e^{-c_2\sqrt{\log x}}$, which is $O_A(xe^{-c\sqrt{\log x}})$. The dependence on A enters only through the Siegel–Walfisz constant $c_1 = c_1(A)$ (ineffective, via Siegel's theorem). □

Remark 11 (Exact local structure)**.** *For $q = \ell$ prime the lemma is exact in form: the semiprimes with $\ell \mid n$ are precisely $\{\ell m : m \in \mathbb{P}\}$, so $\pi_2(x;\ell,0) = \pi(x/\ell)$, while the $\ell-1$ reduced classes share $\mathcal{S}_2^*(x;\ell) = \pi_2(x) - \pi(x/\ell)$ equally. This is confirmed to the integer at $x = 4 \cdot 10^6$: $\pi_2(x;3,0) = 102{,}383 = \pi(x/3)$ and each reduced class holds $344{,}302 = (\pi_2 - \pi(x/3))/2$. The class-$0$ suppression $\pi(x/\ell)/\pi_2(x) \sim 1/(\ell\log\log x)$ is exactly the* $\log\log$ *deficit that produces the channel weight W_f and the blocking of Lemma 1.*

The main term. On a major arc $\alpha = a/q_0 + \beta$ ($(a,q_0)=1$, $|\beta| \le Q/X$) the prime sum is classical, $S_{\mathbb{P}}(\alpha) = \frac{\mu(q_0)}{\phi(q_0)} v_1(\beta) + O(Xe^{-c\sqrt{\log X}})$ with $v_1(\beta) = \int_2^{2X} e(t\beta)\,dt/\log t$. For the semiprime sum, splitting into residues and applying Lemma 6 (and Remark 11 for the non-reduced classes) yields

$$S_{\mathcal{S}_2}(\alpha) = G_2(a,q_0)\,v_2(\beta) + O(Xe^{-c\sqrt{\log X}}), \qquad G_2(a,q_0) = \sum_{b\,(q_0)} e_{q_0}(ab)\,\gamma_{q_0}(b),$$

where $\gamma_{q_0}(b)$ is the local density of semiprimes in class b and $v_2(\beta) = \int e(t\beta)\,d\Pi_2(t)$, Π_2 the smooth semiprime count. For $q_0 = \ell$ a short computation from Remark 11 gives the multiplier $G_2(a,\ell) = \frac{1}{(\ell-1)\pi_2(x)}(\ell\,\pi(x/\ell) - \pi_2(x))$, and pairing it with the prime factor $\mu(\ell)/\phi(\ell) = -1/(\ell-1)$ reproduces, after summation over a and multiplication over q_0, the local factor $(\ell-1)/(\ell-2)$ at primes $\ell \mid N$ removed and the suppression at $\ell = r \mid N$—i.e. the singular series of Lemma 1. Hence

$$\mathrm{main}(N) = \mathfrak{S}_2(N;Q)\,J(N)(1+o(1)), \quad \mathfrak{S}_2(N;Q) = 2C_2\,\mathfrak{S}(N)\,W_f(N) + O(Q^{-1}), \quad J(N) \asymp \frac{X\log\log X}{\log^2 X},$$

the tail $\mathfrak{S}_2(N) - \mathfrak{S}_2(N;Q) \ll Q^{-1}$ being the standard absolutely-convergent completion of the singular series.

Uniformity in the small factor. The per-channel form of Proposition 3 is this same computation carried out *uniformly* for the small prime factor $r \le X^\eta$: for each such r the modulus runs over $q_0 \le Q$ and the local semiprime multiplier γ_{q_0} is controlled uniformly by Lemma 6, while the channels $r > X^\eta$ are absorbed by Lemma 5. Pairing the local product $G_2(a,q_0)$ with the prime factor $\mu(q_0)/\phi(q_0)$ reproduces, at each prime ℓ, the singular factor $(\ell-1)/(\ell-2)$ for $\ell \mid N$ and the suppression at $\ell = r \mid N$—that is, exactly $2C_2\mathfrak{S}(N)\frac{r-1}{r-2}I_r(N)$ per unblocked channel. This is the sense in which the major-arc input of Theorem 3 is uniform over $3 \le r \le \sqrt{N}$.

Remark 12 (Numerical check of the main term)**.** *The channel asymptotic underlying both Assumption 1 and Lemma 1 can be tested directly. Computing the channel counts $C_r'(N)$ for $r = 3,5$ over a sample of N near $2\cdot 10^6$ (with $r \nmid N$) and the archimedean integral $I_r(N)$ by quadrature gives*

$$\frac{C_r'(N)}{2C_2\,\mathfrak{S}(N)\,\frac{r-1}{r-2}\,I_r(N)} = 0.996 \pm 0.004 \quad (r=3), \qquad 0.995 \pm 0.008 \quad (r=5),$$

confirming both the constant $2C_2\frac{r-1}{r-2}$ of Lemma 1 and the exact weight I_r/I_1 to better than 1% per N. (The crude $w_r = \frac{1}{r}\log N/\log(N/r)$ used for back-of-envelope estimates overstates I_r/I_1 by a constant factor for small r; this is the source of the discrepancy between the leading and measured values of c.)

Tracking of A. The three parameters are tied through a single $Q = (\log X)^B$. Lemma 6 and the prime major arcs hold for $q_0 \le Q$ with any fixed B (taking $A = B$); their errors are $O(Xe^{-c(B)\sqrt{\log X}})$, super-polynomially small. The variance (Proposition 2) needs $Q = (\log X)^{2A_0+8}$ to give relative variance $\ll (\log X)^{3-2A_0}$. Choosing $B = 2A_0+8$ makes all three compatible, and the exceptional set in Theorem 3 is then

$$\#\{X < N \le 2X : |R_2(N) - \mathrm{main}(N)| > \varepsilon\,\mathrm{main}(N)\} \ll_\varepsilon X(\log X)^{3-2A_0} + Xe^{-c\sqrt{\log X}} \ll_C \frac{X}{(\log X)^C}$$

for any fixed C (take $A_0 = \frac{C+3}{2}$). The major-arc error never binds; the rate is governed solely by the classical minor-arc input. This completes the bounded reduction of the *linear* almost-all asymptotic $R_2 = \mathrm{main}(1 + o(1))$ (Theorem 3) to standard prime-distribution estimates, at the level of the stated major-arc input; the analytic machinery is classical, and the one ingredient we make explicit is the singular-series cancellation of Lemma 1. The log-regression statement $\beta_2(N) \to 1$ then follows from the lower-tail hypothesis of Corollary 1, the one ingredient we do not prove.

B Cross-disciplinary reframings (summary)

The Goldbach–Chen frontier can be recast in the languages of statistical physics, network science, markets, simplicial topology, control, information theory, and discrete geometry. We are explicit that these are *not* independent findings: in each the governing quantity is the same singular series $\mathfrak{S}(N)$, so they are one fact—$\mathfrak{S}$ sets the local densities—seen in several mirrors. We summarize them here and develop them, with their figures, in the accompanying supplement; the honest negatives are reported as such.

- **Arithmetic ground-state energy.** With energy $H(a, b) = \Omega(a) + \Omega(b)$, Goldbach is "$E(N) = \min H = 2$"; the ground-state degeneracy is R_1 and the first excited level $2R_2$. The partition function $Z_N(\beta) = (f_\beta * f_\beta)(N)$ costs one FFT per β, and the melting temperature $\beta^\star(N)$ at which Goldbach order localises is an almost deterministic function of the singular series ($\mathrm{corr}(\beta^\star, \log \mathfrak{S}) = -0.92$).
- **Adversarial robustness.** The Goldbach graph is *random-robust but structure-fragile*: deleting 95% of primes at random breaks no N in the window, yet deleting one reduced class mod 3 (half the primes) destroys 59%—a manufactured local obstruction.
- **Liquidity market.** Reading N as a market cleared by two prime inputs, the Chen semiprime substitute is *fair-weather liquidity*: under a supply shock the substitute spread decays linearly, $S_C(\phi) = S_C(0)(1 - \phi) \to 0$, with a negligible resilience premium.
- **Simplicial topology (negative).** For the *binary* problem the additive-complexity filtration is a disjoint matching—its 0-persistence is just the energy spectrum and higher homology vanishes; genuine higher structure appears only at the ternary level, which is anti-clustered. A reported negative.
- **Control / value of information.** A residue-aware search ordering—the one reframing that yields a concrete algorithm—cuts the expected per-N Goldbach-search cost from 5.5 to 2.5 tries at budget $B = 13$, the gain per prime tracking $(\ell - 1)/(\ell - 2)$.
- **Information content.** Conditioning $\log R_1$ on N mod 30030 leaves a residual of 0.11% that is white noise: up to that modulus the comet is almost entirely a singular-series codeword (a statement relative to the conditioning set, not a theorem).
- **Discrete geometry of $p + rs = N$.** The Chen solutions of a typical N concentrate against the Goldbach edge (median Li–Liu exponent $a = 1.30$, far below the 1.9 threshold), the geometric form of the residual-exponent and factor-balance findings of the body.

The full development, with the corresponding figures (energy, robustness, market, control, information, geometry), is in the supplement `supplement.tex`.

Data and code availability

Every figure, table, and quoted number in this paper is reproduced by the accompanying open-source repository (Python; `numpy/scipy/matplotlib`, with `gurobipy` only for the optional MIP of §7.5). Each `src/` module carries a self-test (`python3 src/<module>.py`) and each figure is regenerated by its `experiments/run_*.py` driver; the central results use `run_counts.py` (FFT counts to $2 \cdot 10^6$), `run_scaling.py` (block-convolution sieve to 10^9), `run_betalaw.py` (the law (25)), `run_estimator_audit.py` (the estimator reconciliation and the $3 \mid N$ atom of §7.2), and `run_exceptional.py` (the concentration of Table 6). Derived tables and the sieve seeds are stored in `data/`. The repository is available at `https://github.com/MathiasRodriguezCastro/goldbach-chen-frontier`, and a versioned snapshot is archived on Zenodo, doi:10.5281/zenodo.20725701 (citation metadata in `CITATION.cff`).

References

[1] Chen, J. R. (1973). On the representation of a larger even integer as the sum of a prime and the product of at most two primes. *Scientia Sinica*, 16, 157–176.

[2] Halberstam, H. and Richert, H.-E. (1974). *Sieve Methods*. Academic Press, London. (London Mathematical Society Monographs, 4.)

[3] Hardy, G. H. and Littlewood, J. E. (1923). Some problems of 'Partitio Numerorum'; III: On the expression of a number as a sum of primes. *Acta Mathematica*, 44, 1–70.

[4] Helfgott, H. A. (2014). The ternary Goldbach problem. *arXiv preprint arXiv:1404.2224*.

[5] Landau, E. (1909). *Handbuch der Lehre von der Verteilung der Primzahlen*. Teubner.

[6] Li, J. and Liu, J. (2026). Theorem $(1 + 1.9)$ on the Goldbach Conjecture. *arXiv preprint arXiv:2606.05224*.

[7] Montgomery, H. L. and Vaughan, R. C. (1975). The exceptional set in Goldbach's problem. *Acta Arithmetica*, 27, 353–370.

[8] Nathanson, M. B. (1996). *Additive Number Theory: The Classical Bases*. Springer.

[9] Oliveira e Silva, T., Herzog, S., and Pardi, S. (2014). Empirical verification of the even Goldbach conjecture and computation of prime gaps up to $4 \cdot 10^{18}$. *Mathematics of Computation*, 83(288), 2033–2060.

[10] Sathé, L. G. (1953). On a problem of Hardy on the distribution of integers having a given number of prime factors. *Journal of the Indian Mathematical Society*, 17, 63–141.

[11] Selberg, A. (1954). Note on a paper by L. G. Sathé. *Journal of the Indian Mathematical Society*, 18, 83–87.

[12] Tenenbaum, G. (2015). *Introduction to Analytic and Probabilistic Number Theory*. American Mathematical Society, 3rd edition. (Graduate Studies in Mathematics 163.)

[13] Vaughan, R. C. (1997). *The Hardy–Littlewood Method*. Cambridge University Press, 2nd edition.